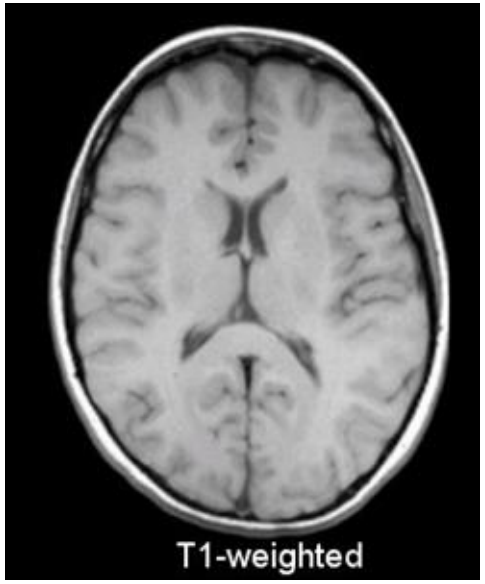
Neuroanatomy for Lesion Tracing & Imaging

Neural Plasticity and Neurorehabilitation Lab

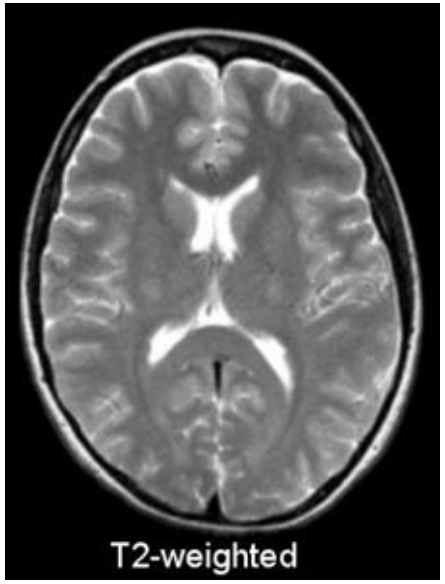


Points to remember:

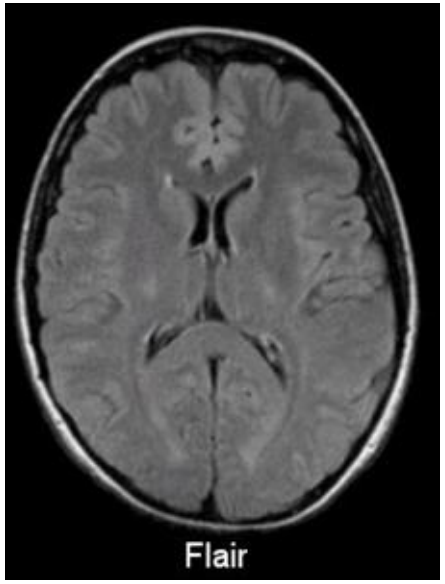
In T1-weighted images, white matter is lighter than gray matter, & CSF is dark.
Conversely, in T2-weighted images, white matter is darker than gray matter, & CSF is bright & prominent.
In both, air & dense bone, which contain relatively few hydrogen nuclei, are dark.



- Gray matter = dark
- White matter = light
- CSF = black
- Lesion = black



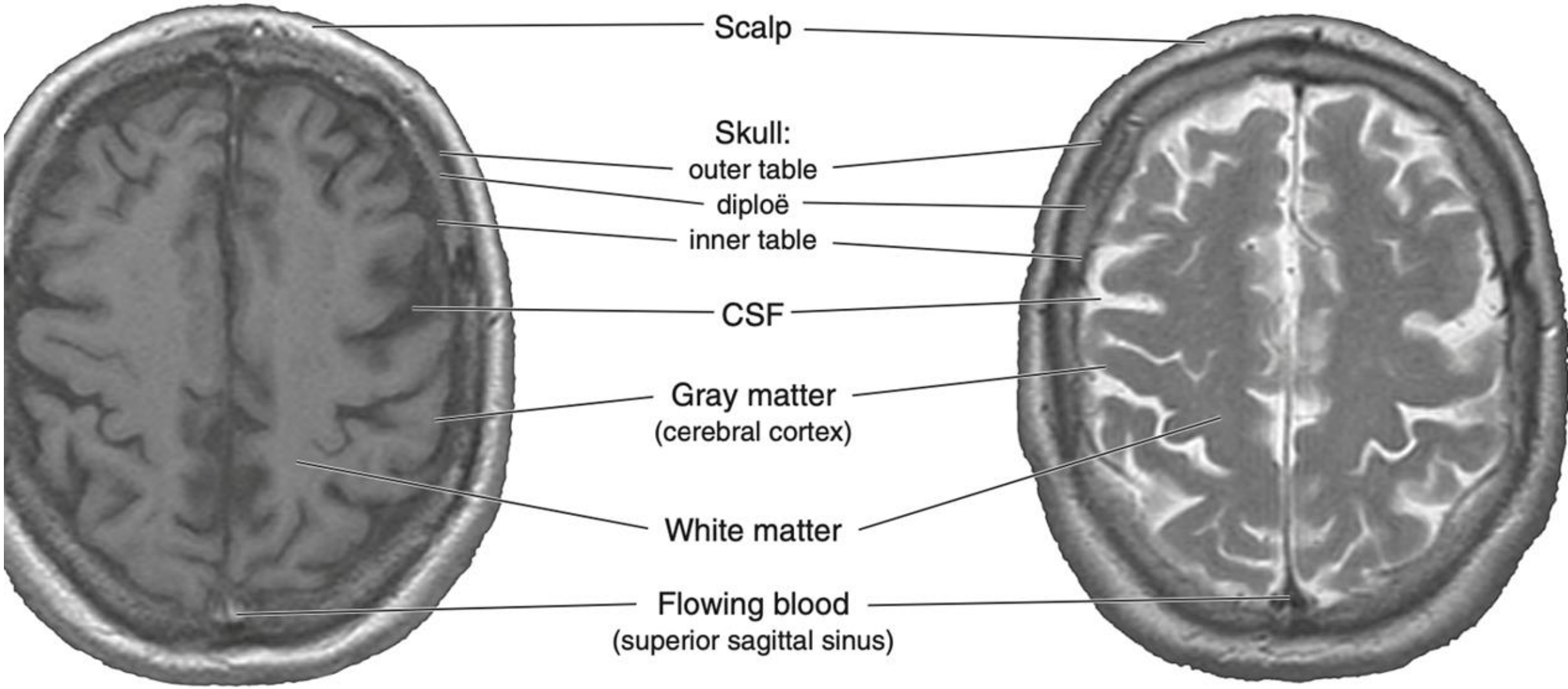
- Gray matter = light
- White matter = dark
- CSF = white
- Lesion = white



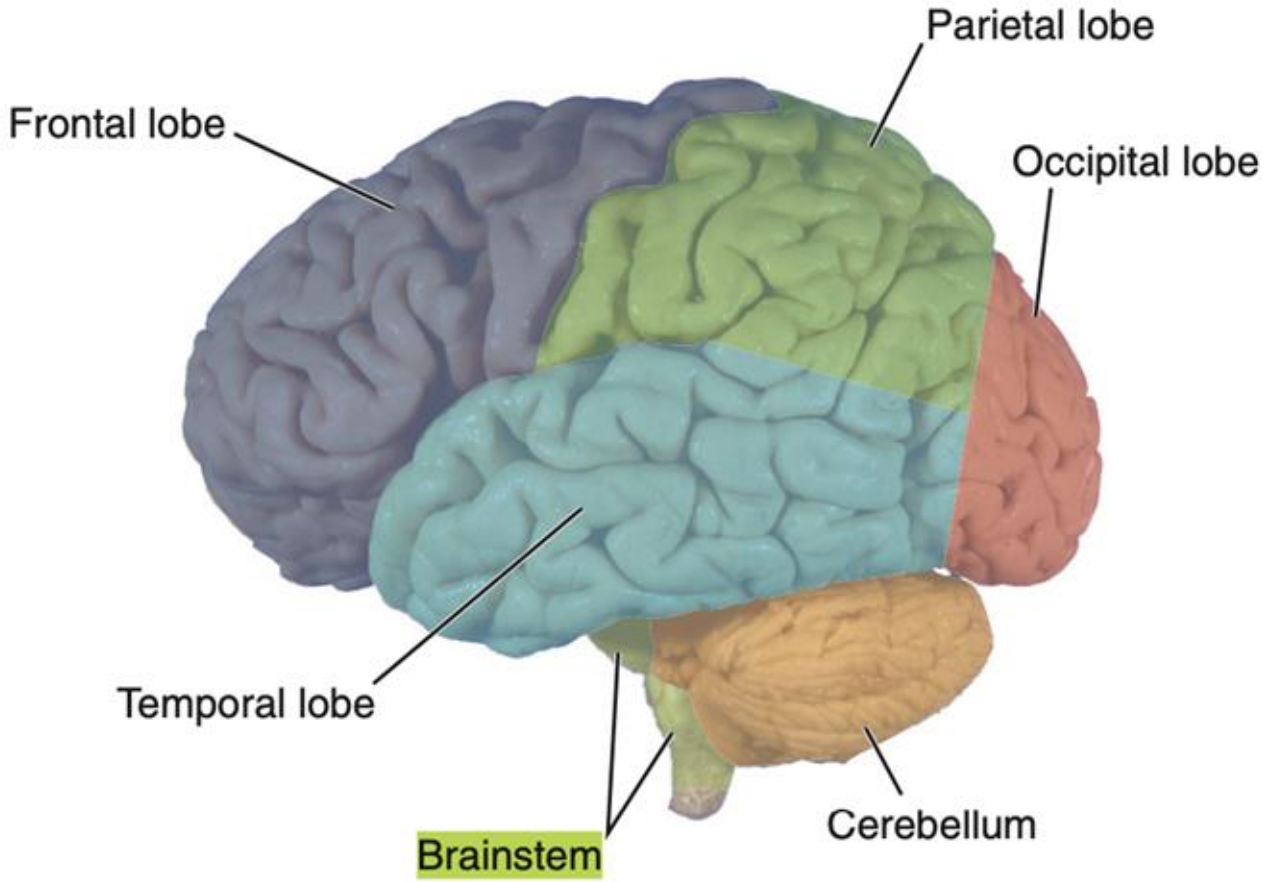
- Gray matter = light
- White matter = dark
- CSF = dark
- Lesion = white

T1

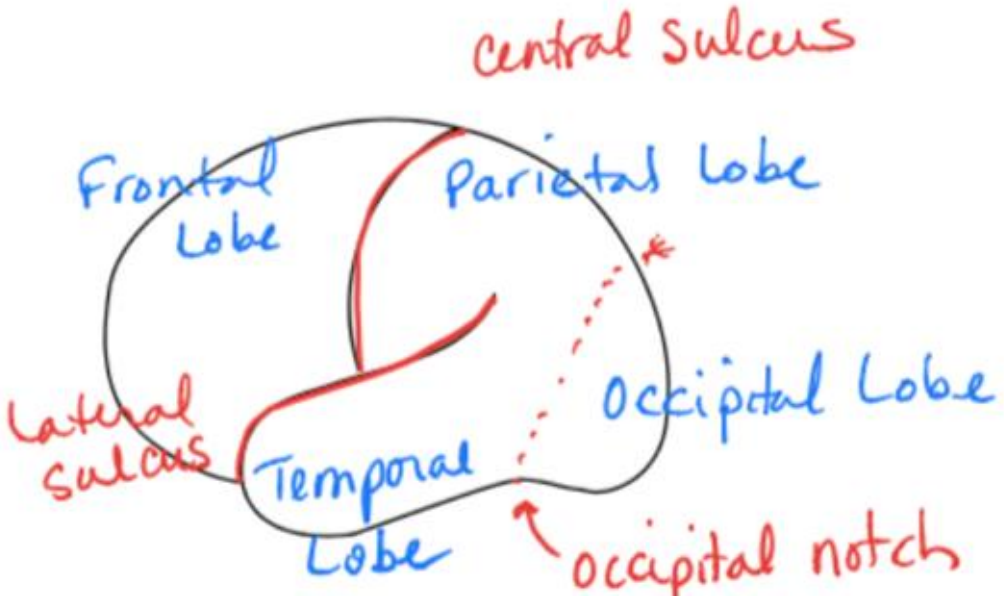
T2

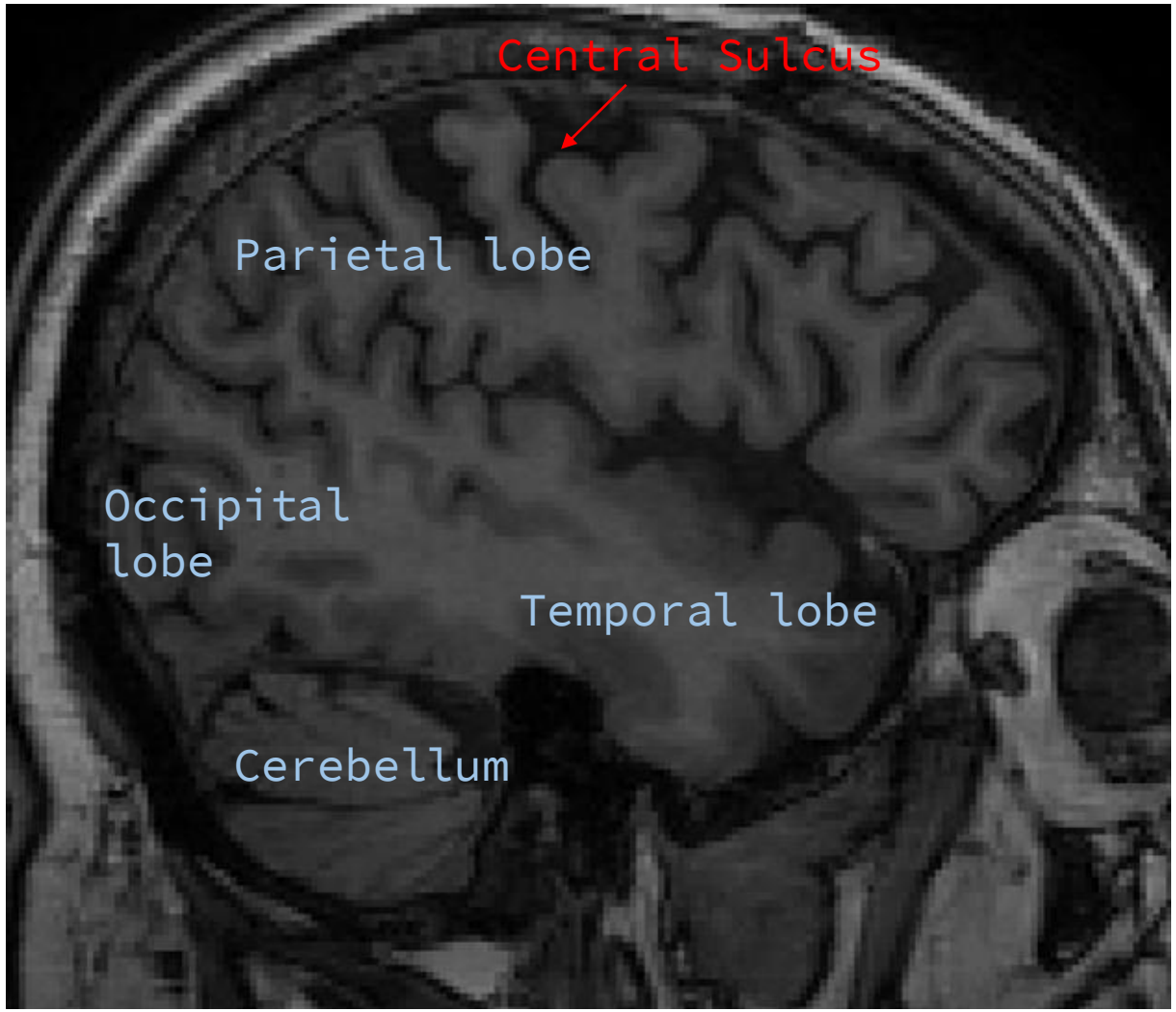

Scalp
Skull:
outer table
diploë
inner table
CSF
Gray matter
(cerebral cortex)
White matter
Flowing blood
(superior sagittal sinus)

Lobes of the brain



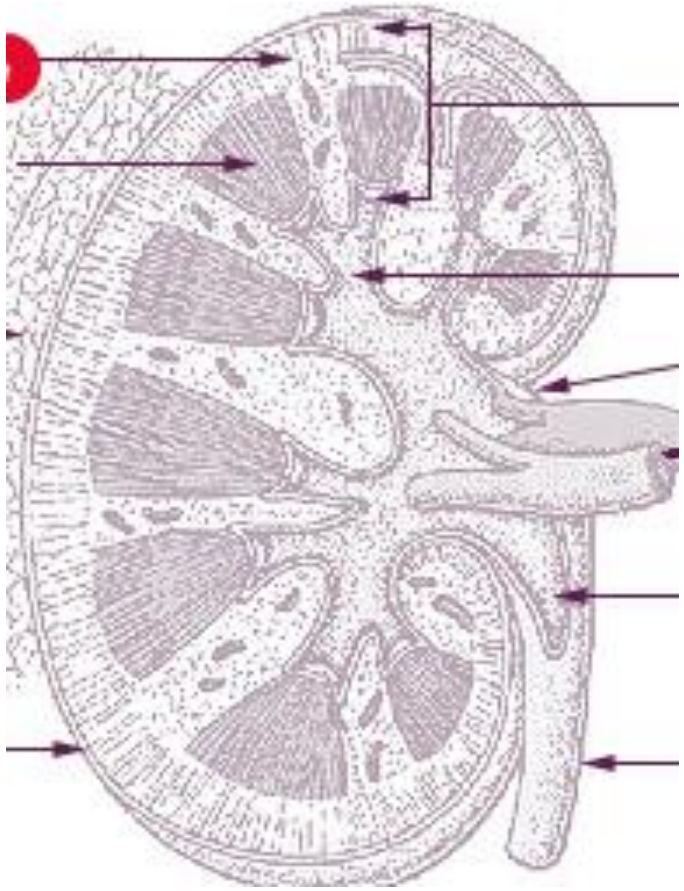
Cortical Lobes



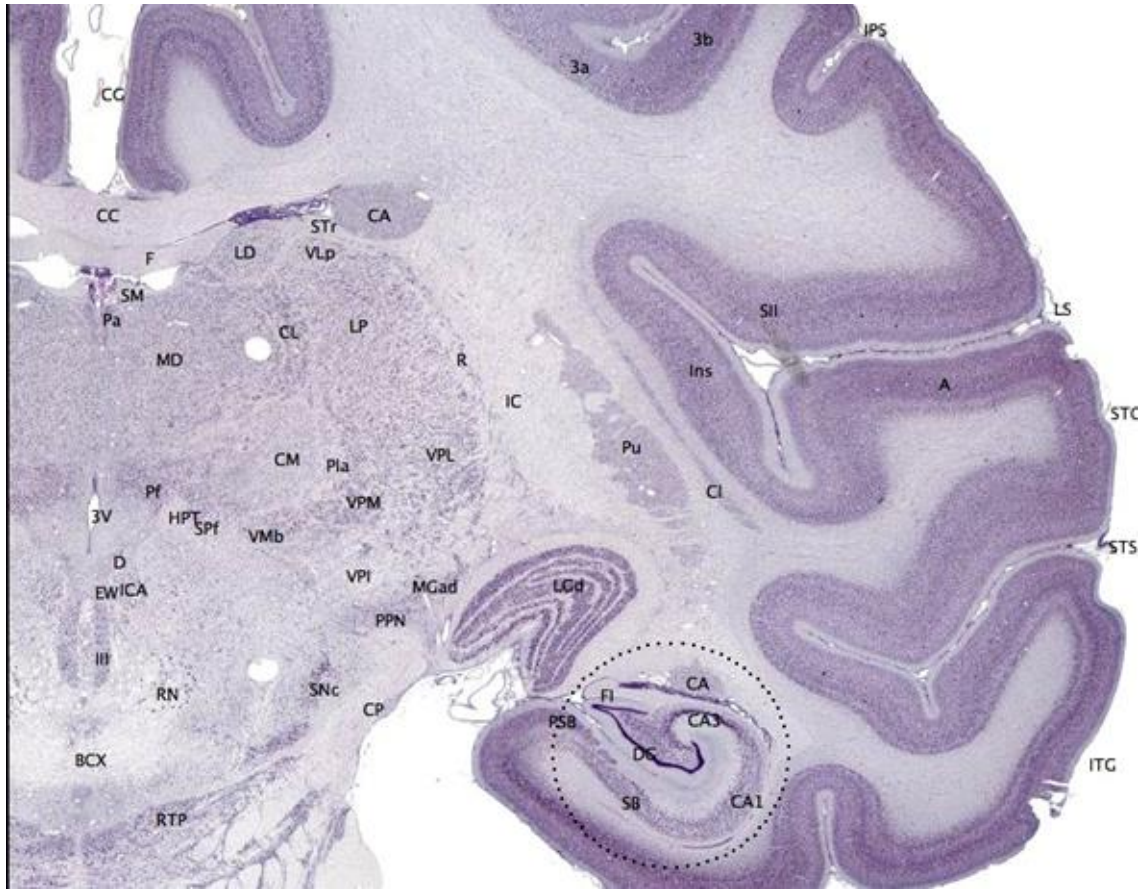


Cortex- Outermost Layer of an Organ

Lots of organs have well defined cortical layers

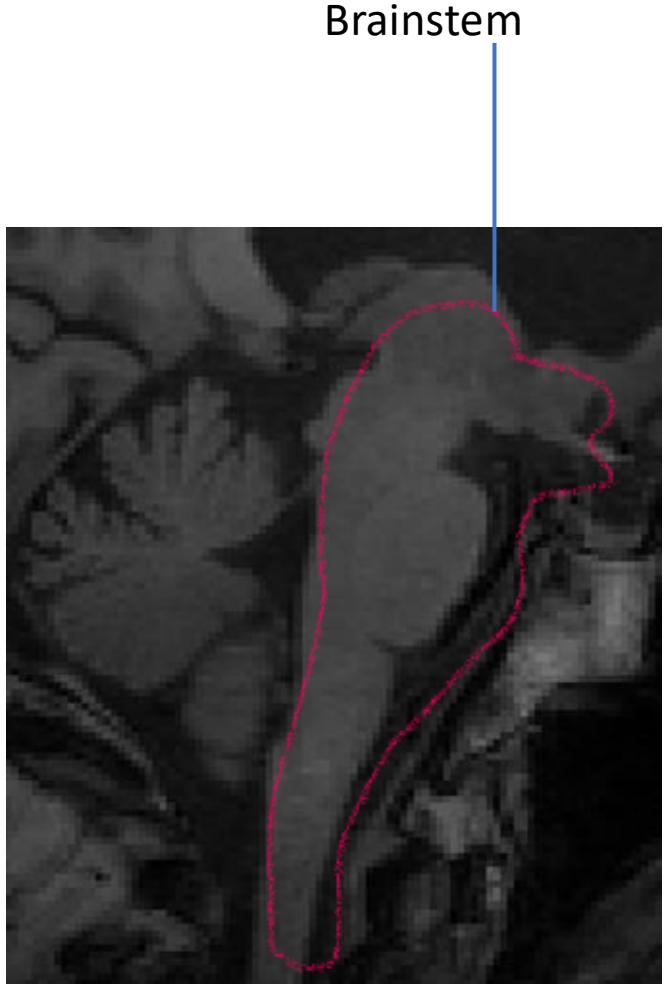


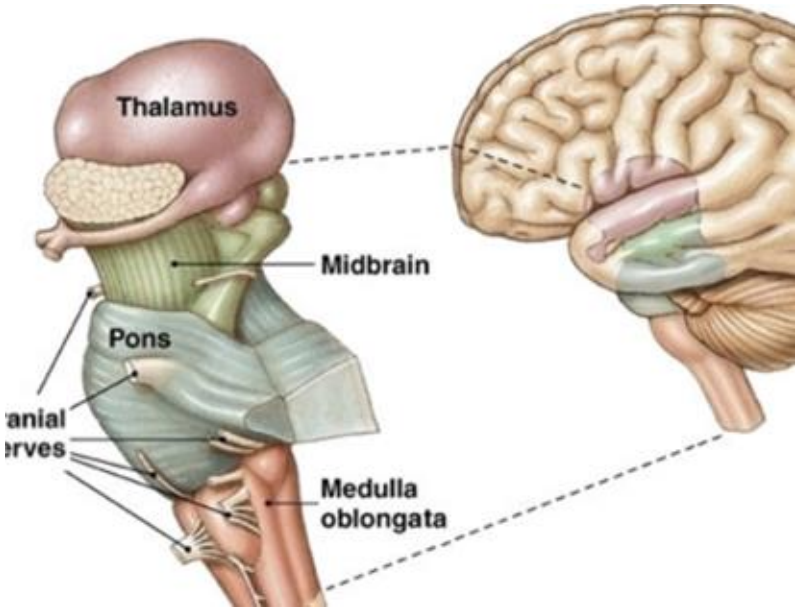
Kidney

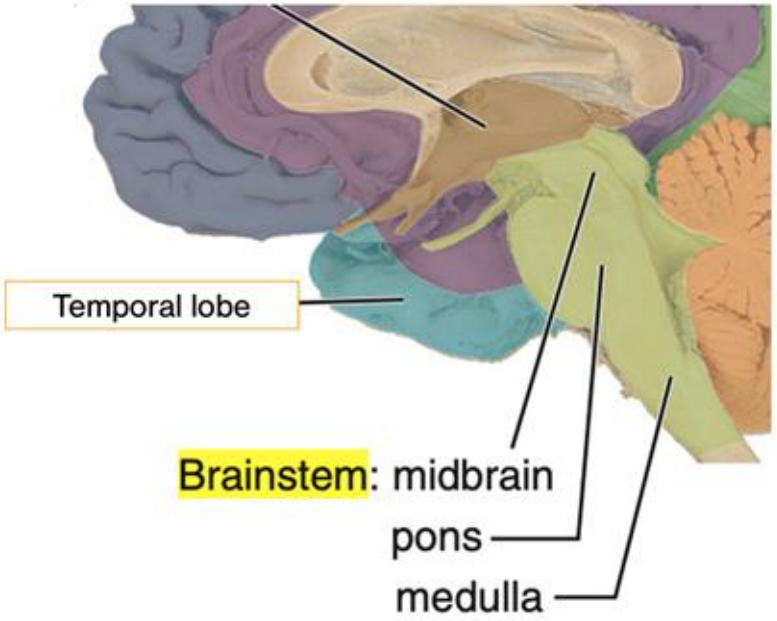


Cerebral Cortex

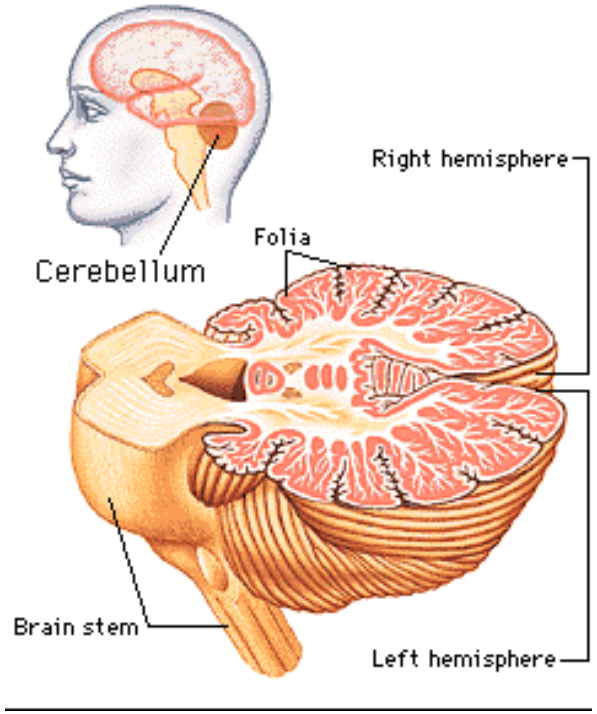
Brainstem

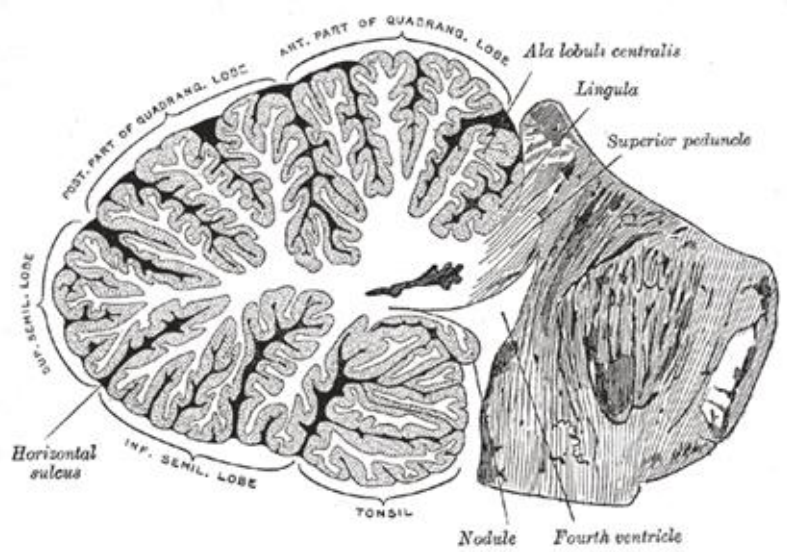

Brainstem

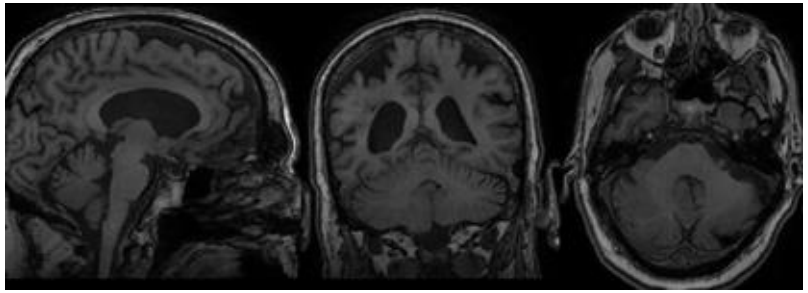

Thalamus
Midbrain
Pons
Medulla
oblongata


Temporal lobe
Brainstem: midbrain
pons
medulla

Cerebellum

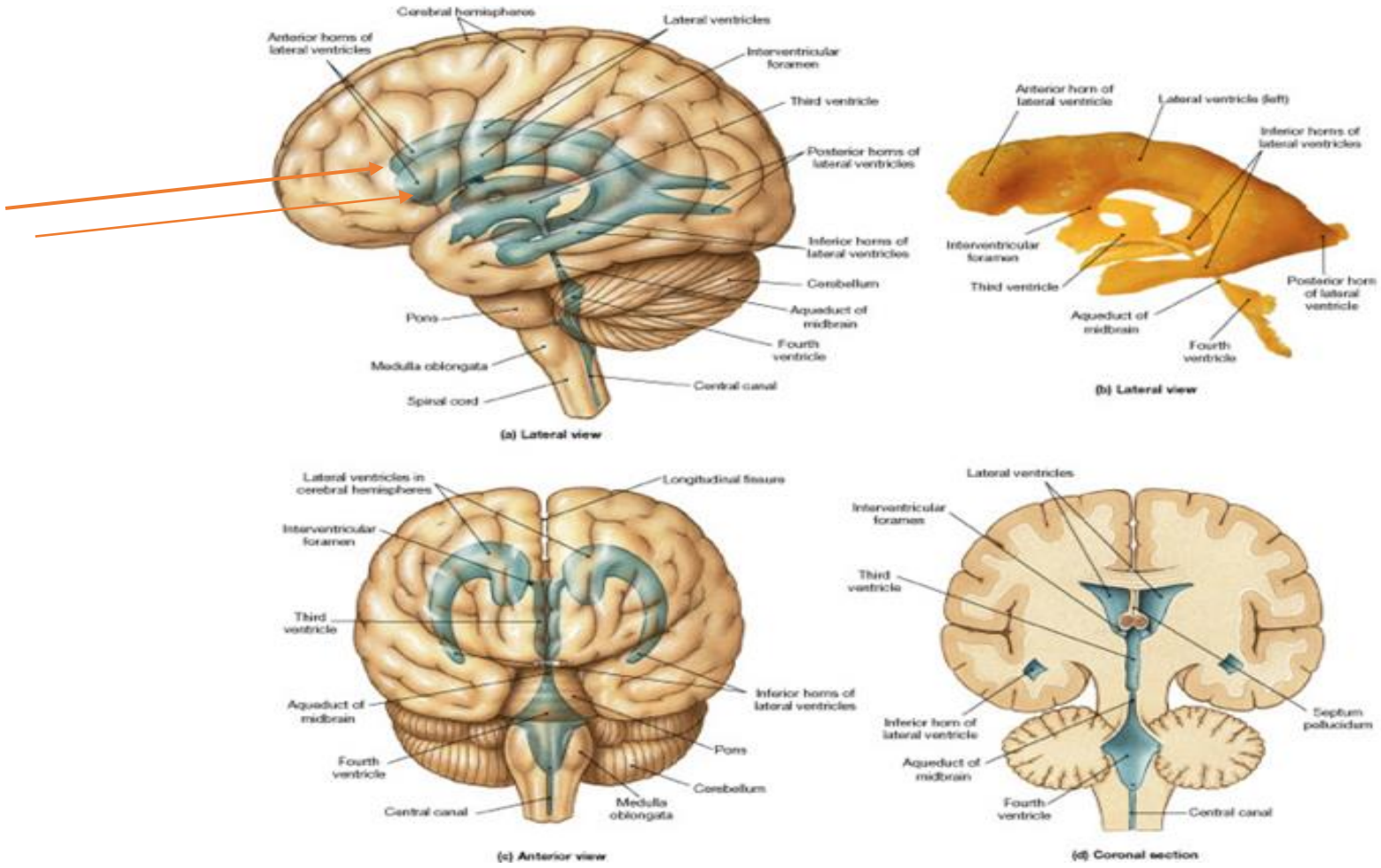






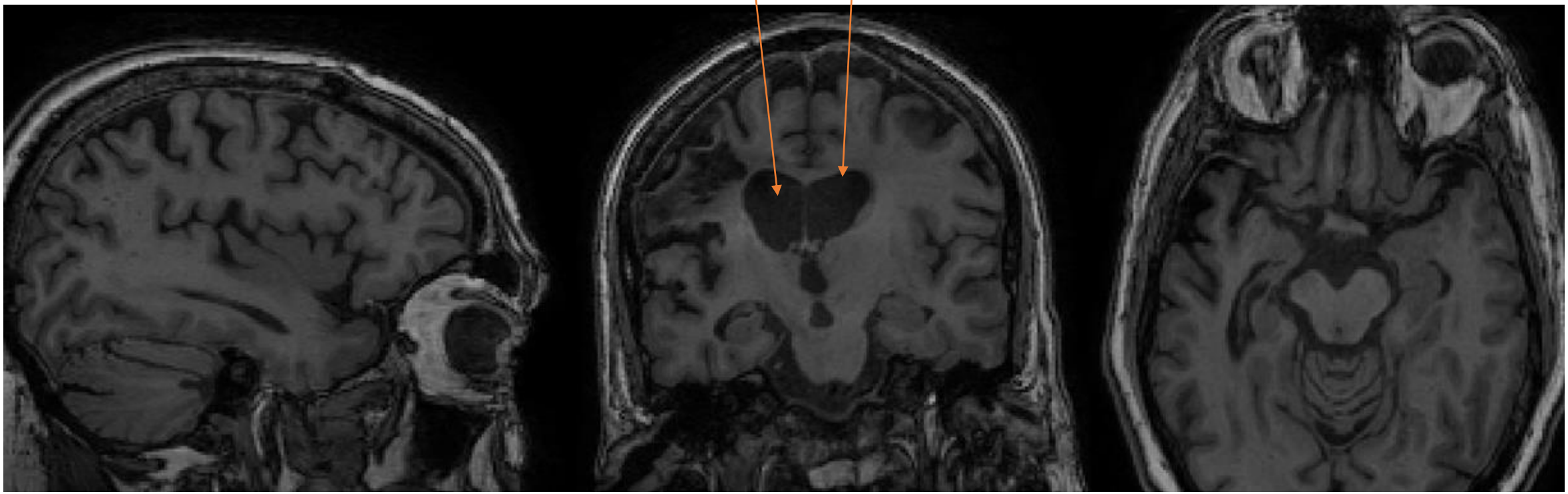
Different ways of viewing the cerebellum

Ventricles


Cerebral hemispheres
Lateral ventricles
Anterior horns of lateral ventricles
Interventricular foramen
Third ventricle
Posterior horns of lateral ventricles
Inferior horns of lateral ventricles
Cerebellum
Aqueduct of midbrain
Pons
Fourth ventricle
Medulla oblongata
Central canal
Spinal cord
(a) Lateral view
Anterior horn of lateral ventricle
Lateral ventricle (left)
Inferior horns of lateral ventricles
Interventricular foramen
Third ventricle
Posterior horn of lateral ventricle
Aqueduct of midbrain
Fourth ventricle
(b) Lateral view
Lateral ventricles in cerebral hemispheres
Longitudinal fissure
Interventricular foramen
Third ventricle
Aqueduct of midbrain
Inferior horns of lateral ventricles
Pons
Fourth ventricle
Cerebellum
Central canal
Medulla oblongata
(c) Anterior view
Lateral ventricles
Interventricular foramen
Third ventricle
Septum pellucidum
Inferior horn of lateral ventricle
Aqueduct of midbrain
Fourth ventricle
Central canal
(d) Coronal section

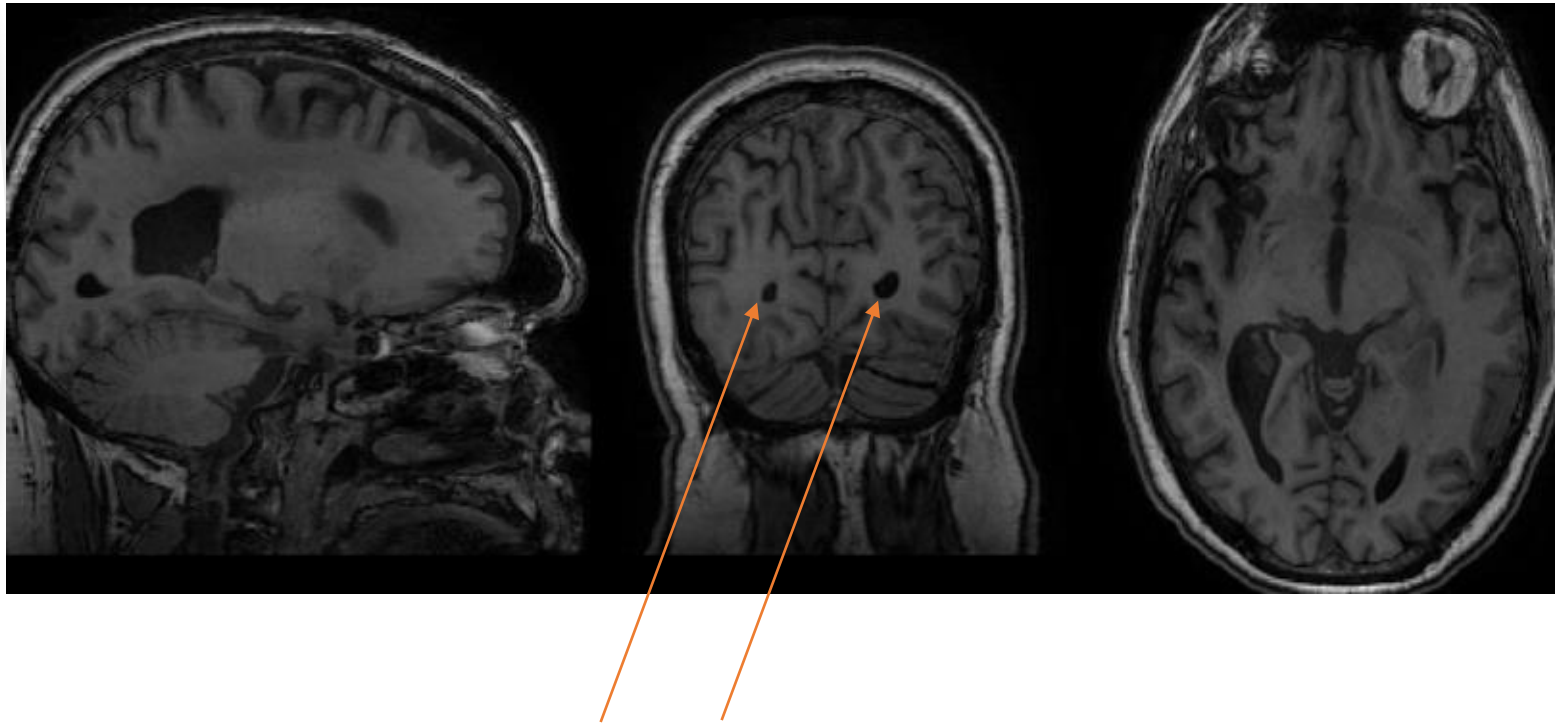
Ventricles Continued

- A nearby lesion will impact the ventricles - in this subject, you can see ventricle atrophy next to the lesion



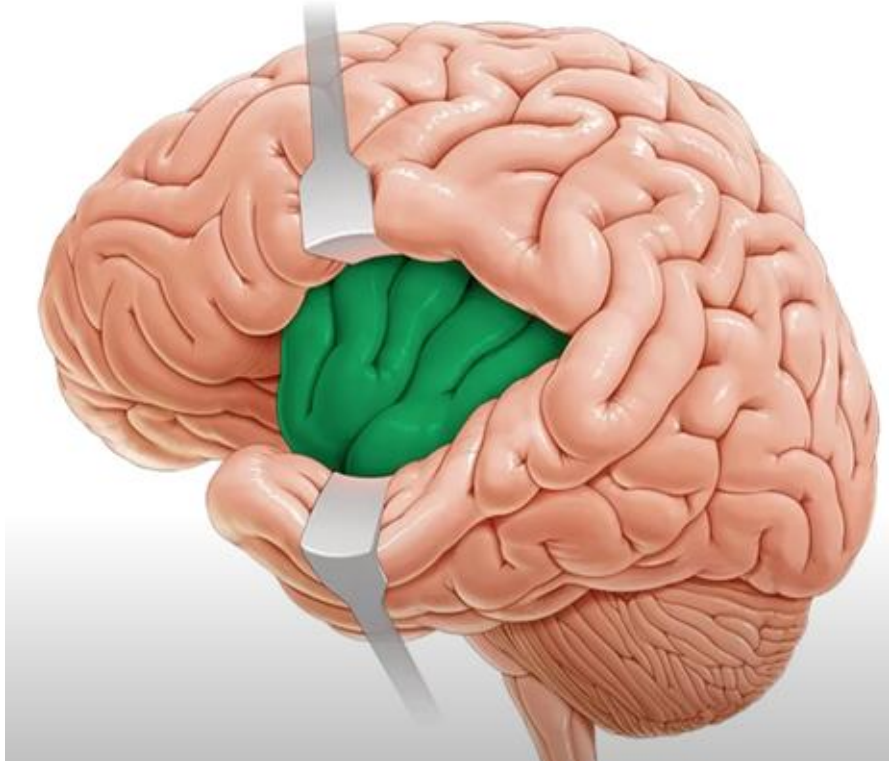
Ventricles

- Pay close attention to head tilt in the scan that might cause the ventricles to look asymmetrical, even if they’re not

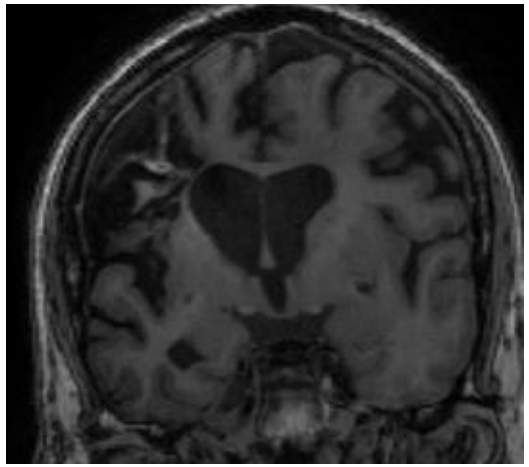


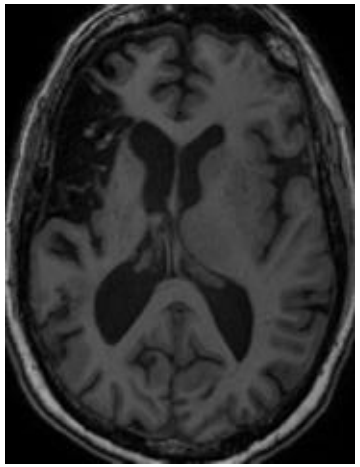
Insula- an important confusing Cortical Region

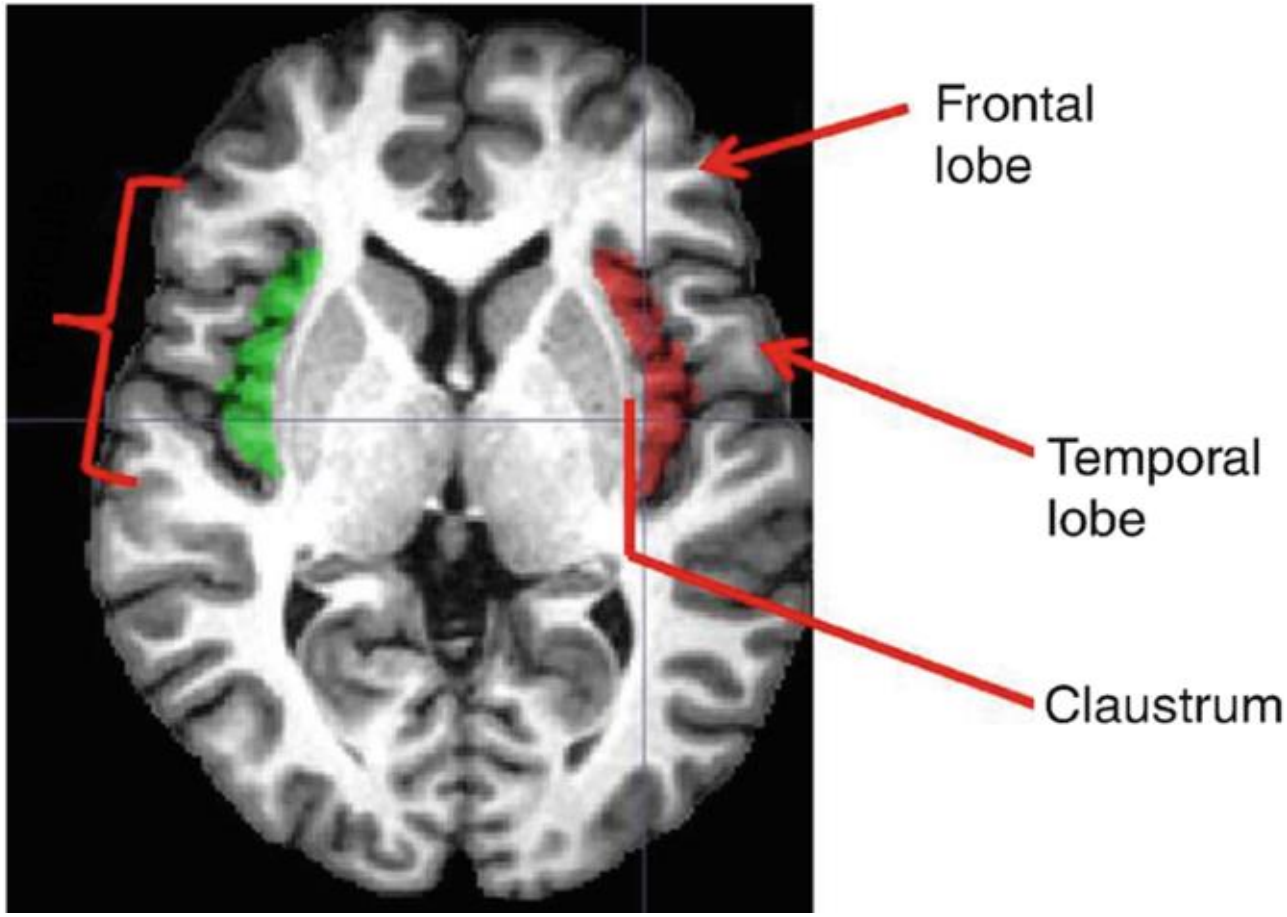




Insula continued

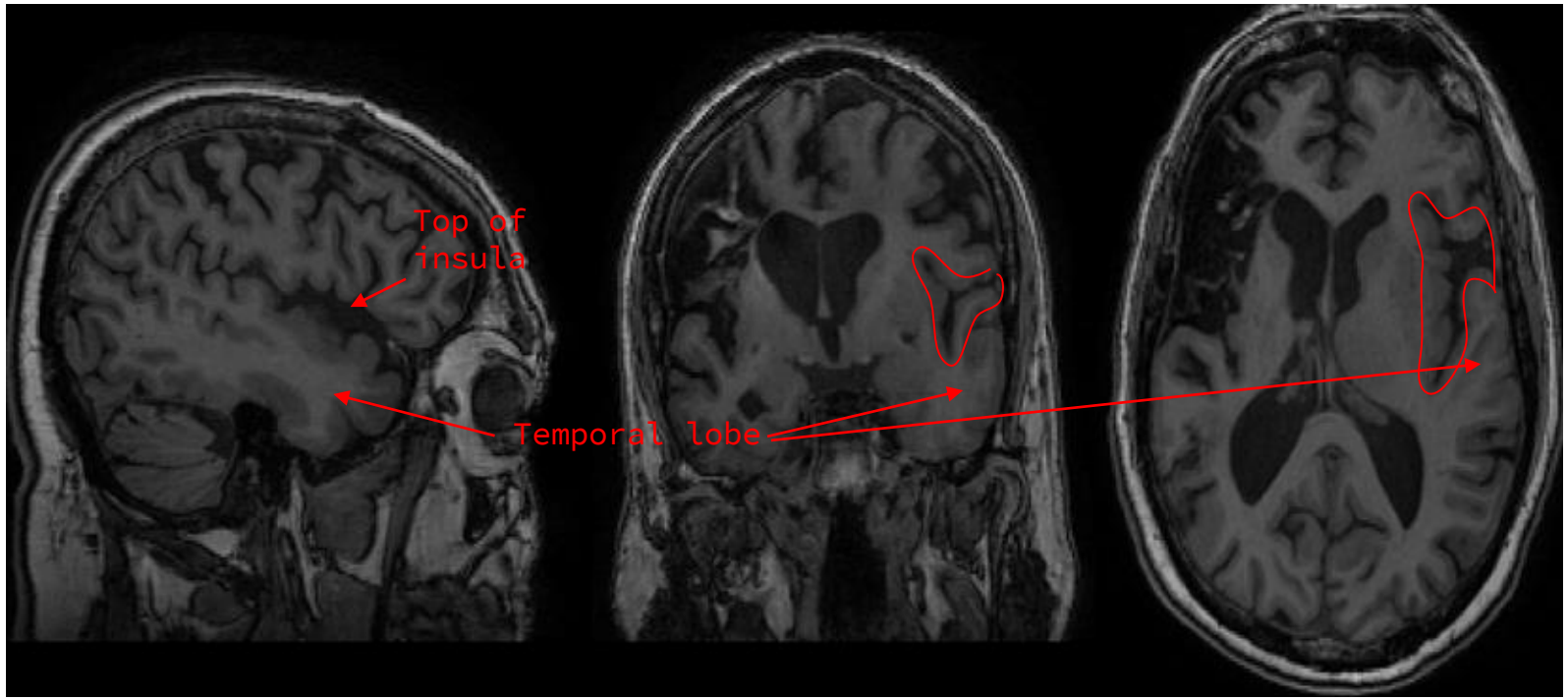






Multiple views of the Insula

- Outlines show the insula, an important region where strokes can occur. It's visualized when the temporal lobe is exposed.

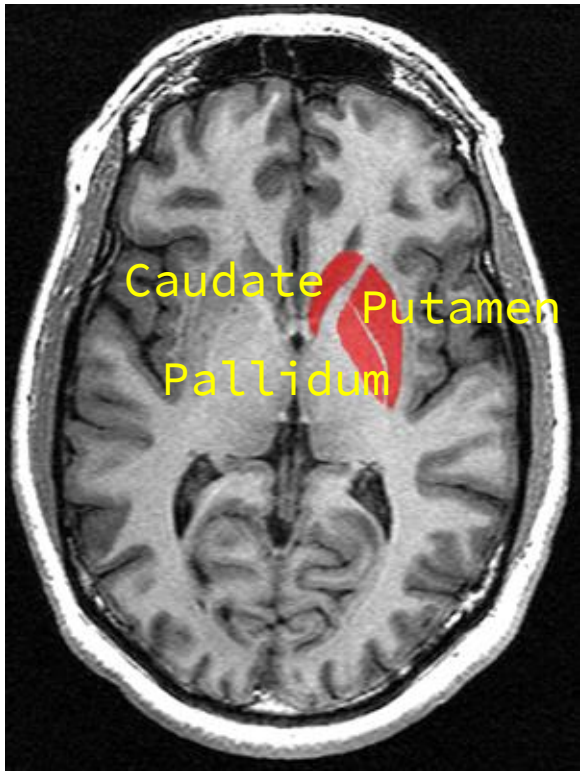


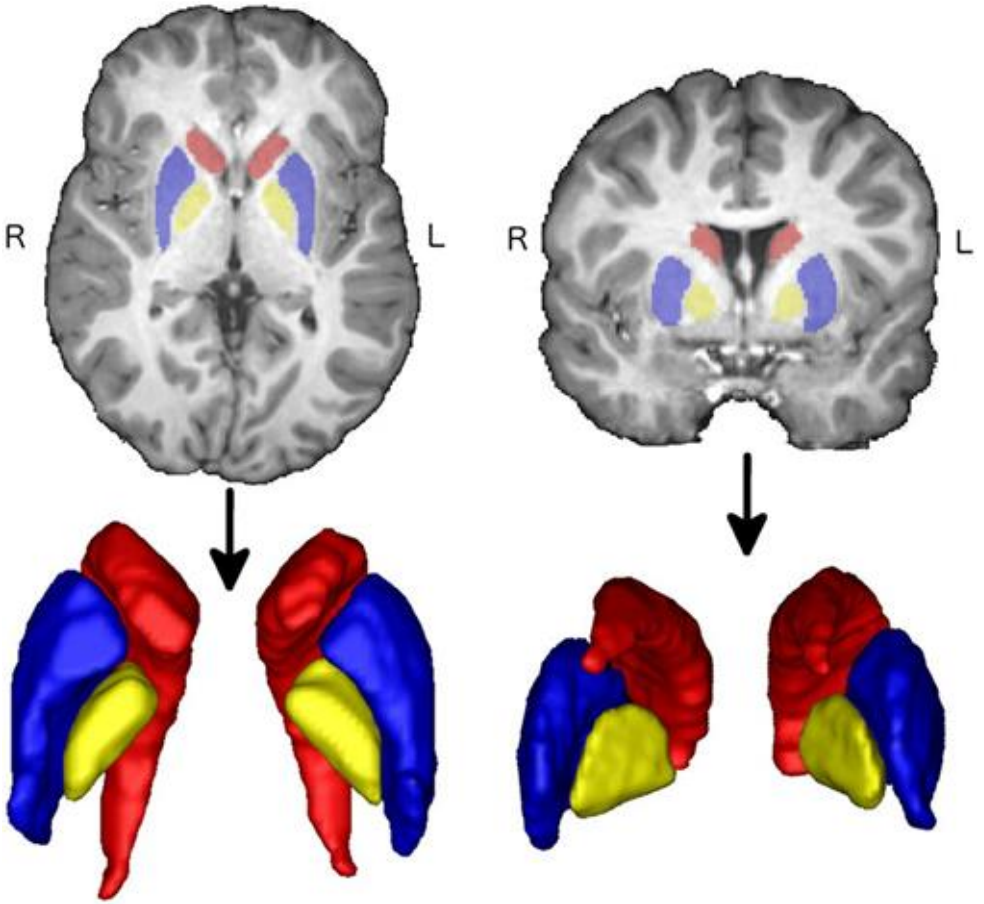
Basal Ganglia - an important gray matter region

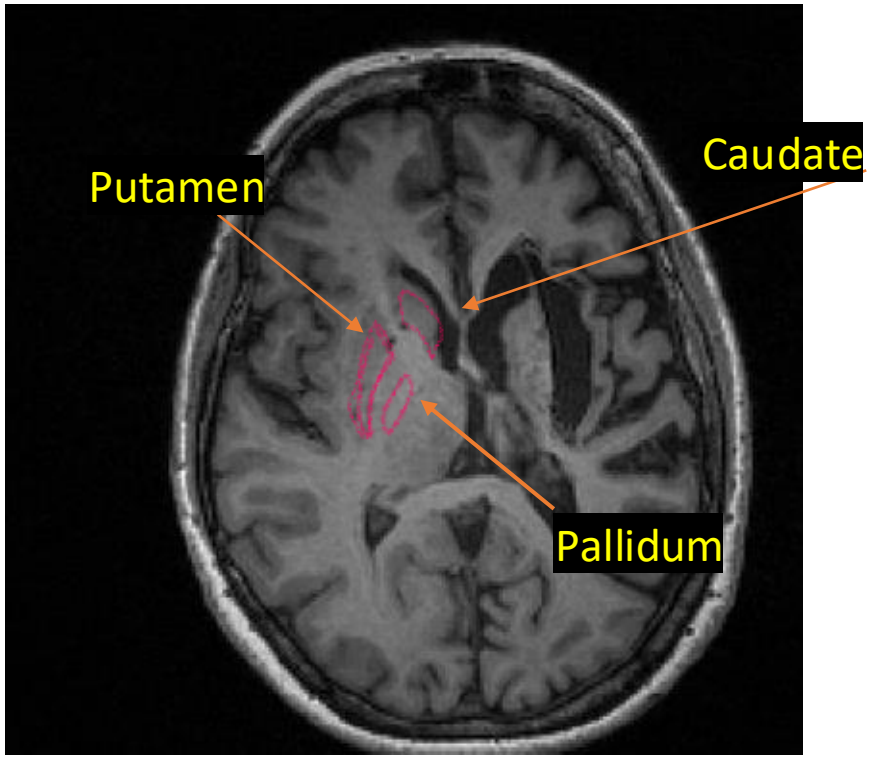
Gray matter → appears darker on T1

Consists of caudate, putamen, pallidum

Lesions will often affect the entire putamen, or run alongside parts of the basal ganglia



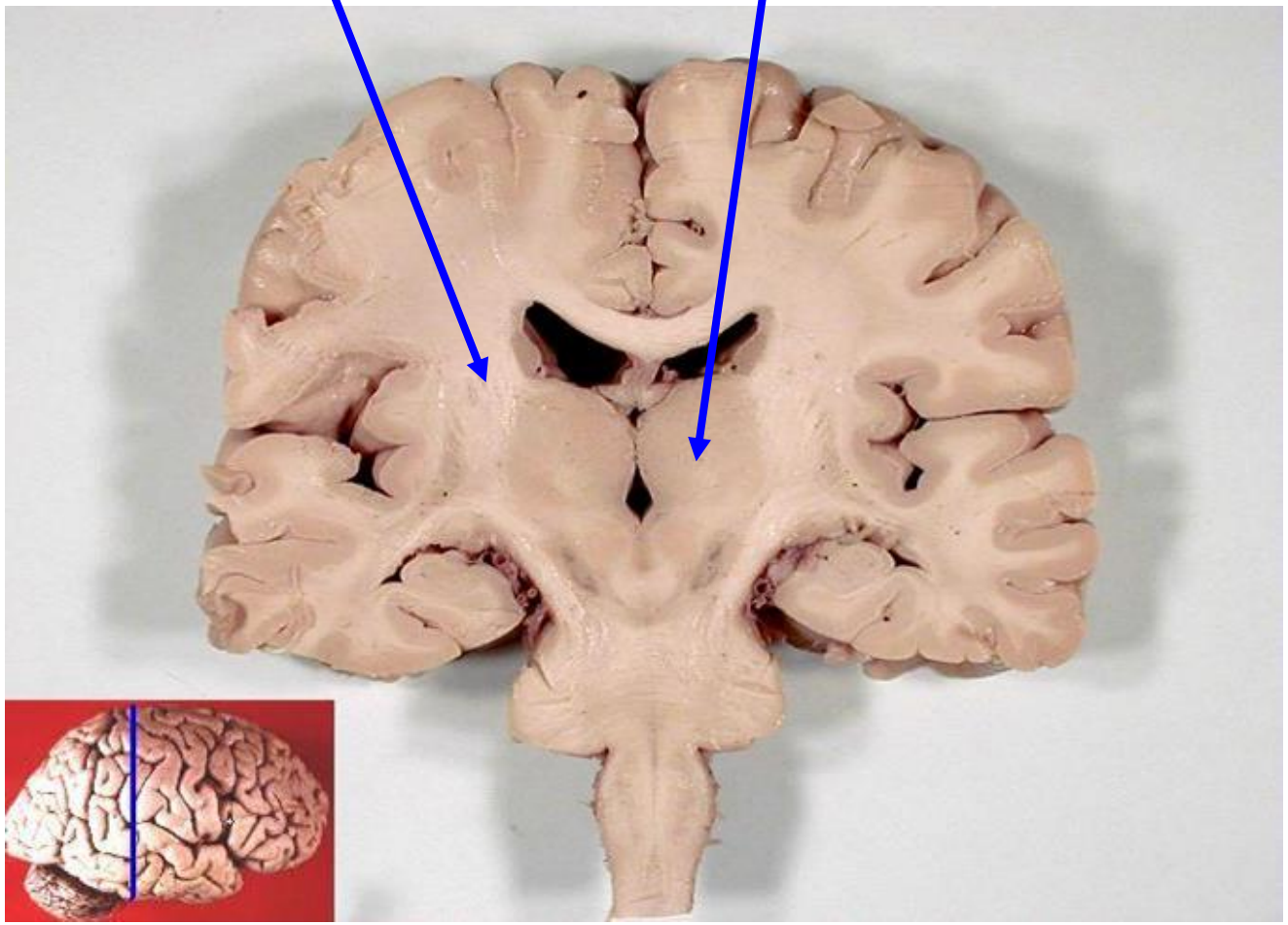




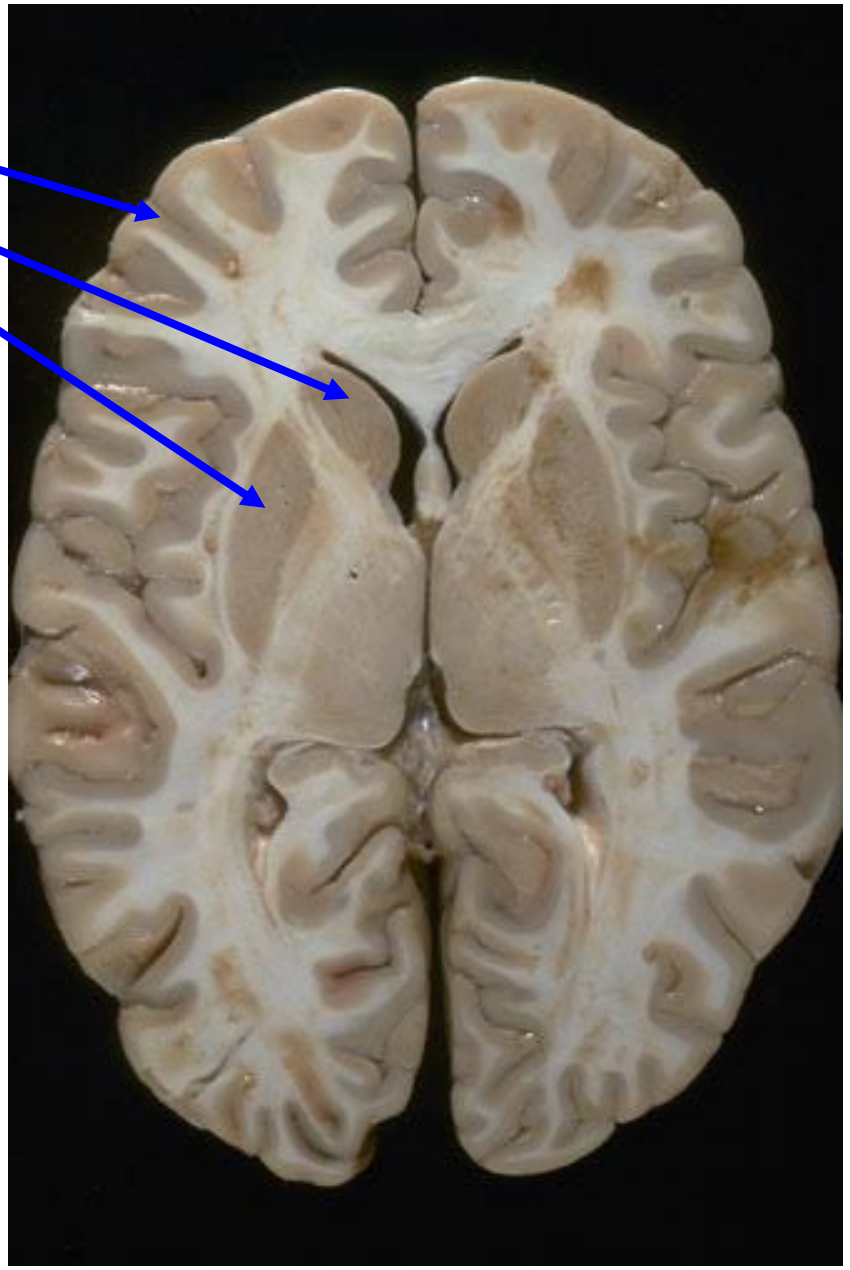
Actual Brain Specimens

White Matter

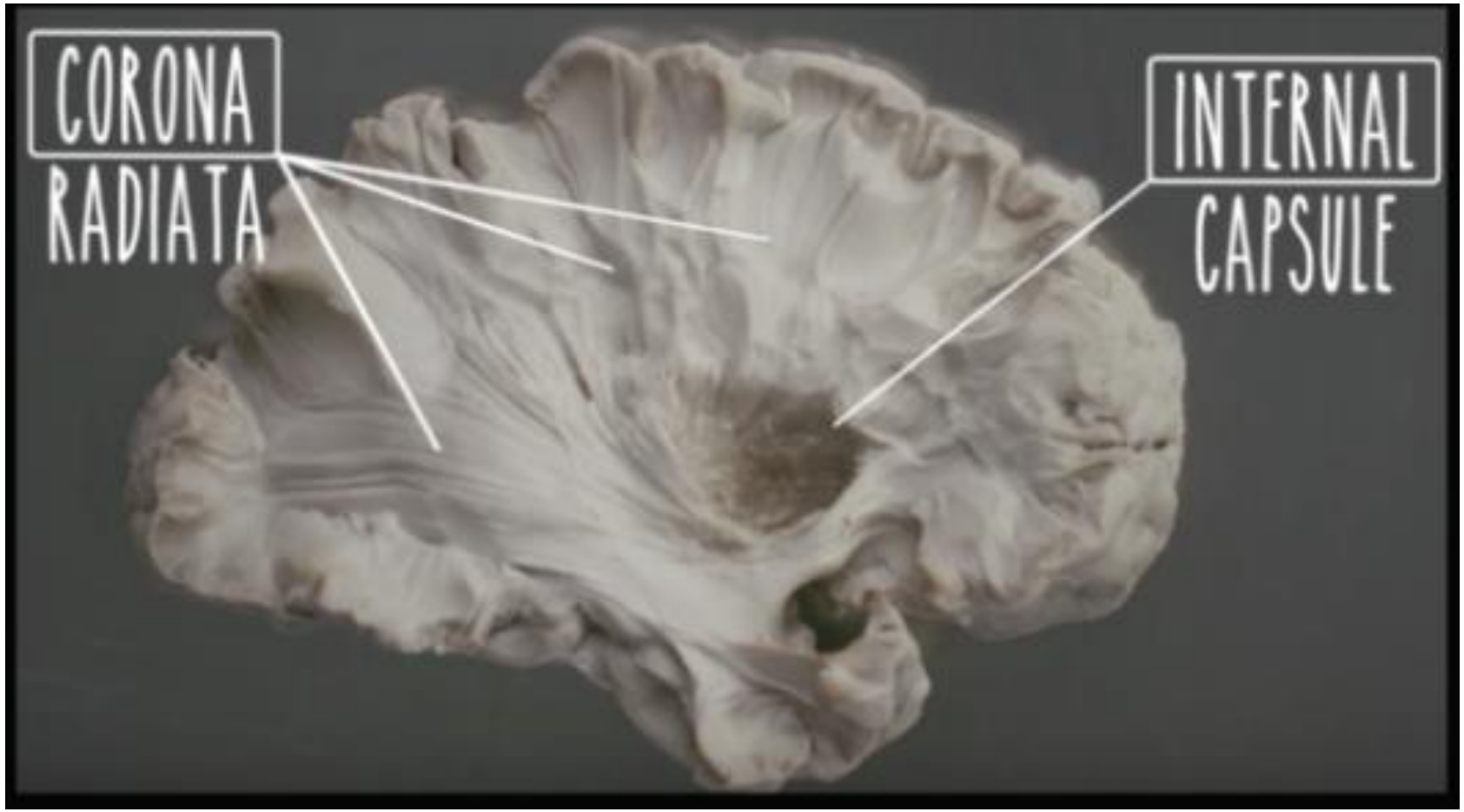
Mostly gray but also some white!



Gray Matter



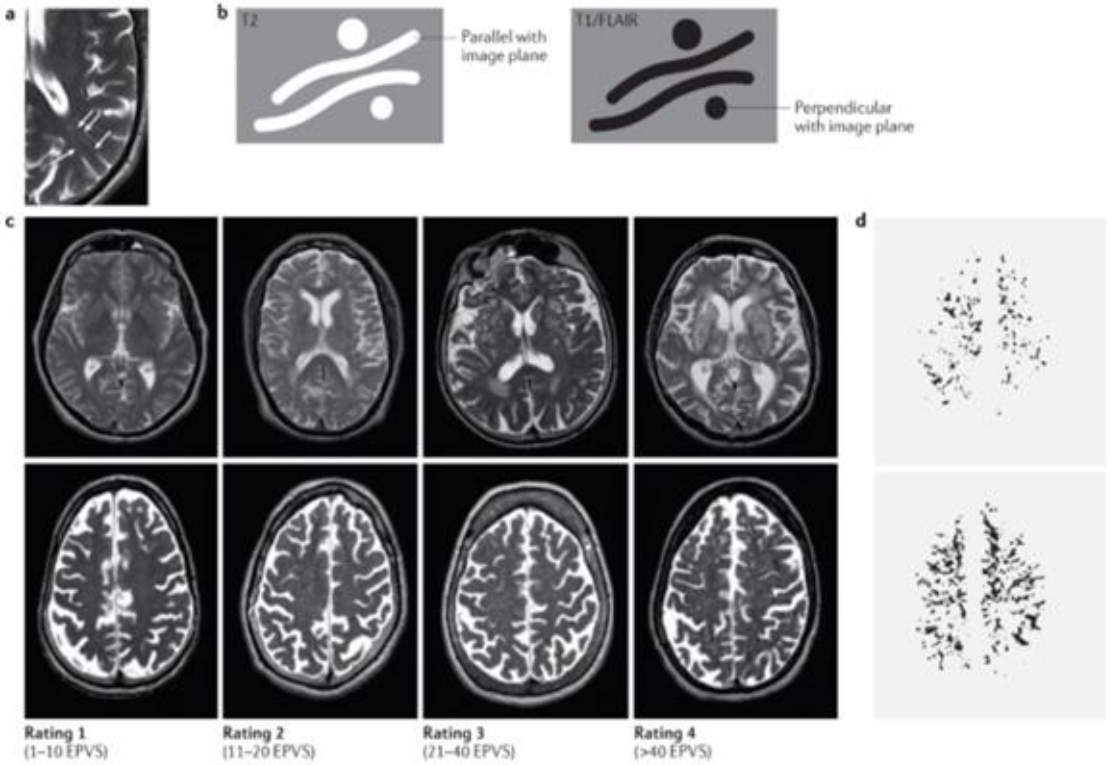
White Matter actually looks like little pathways



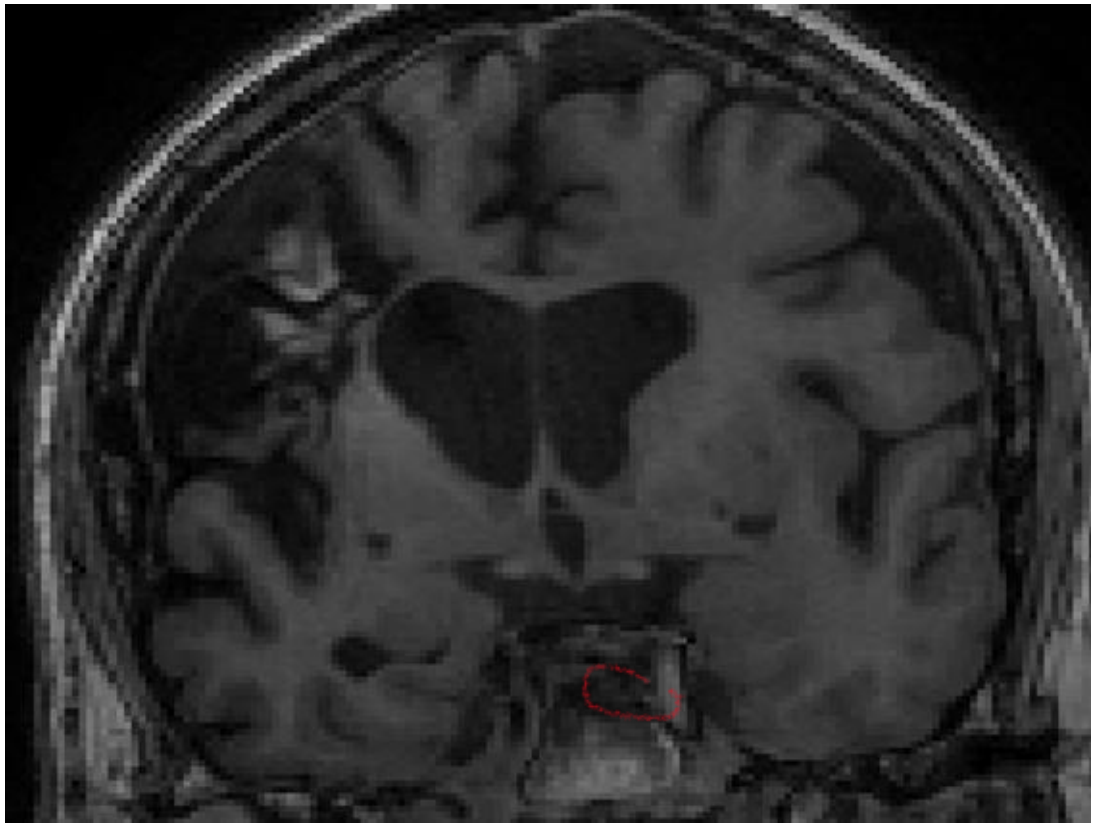
Perivascular Spaces (PVS)

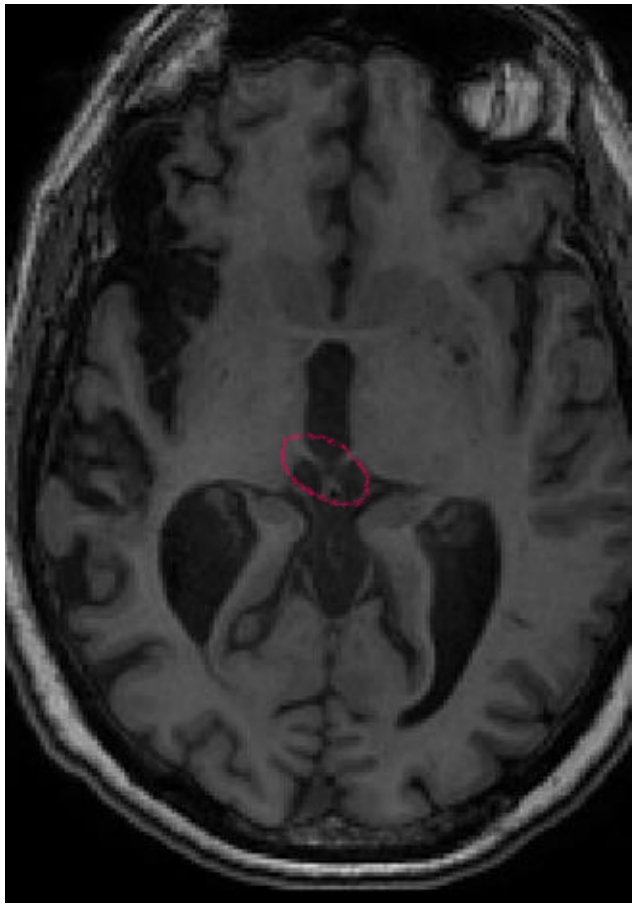
- PVS → surround blood vessels in the brain
- May increase in size with age (thus enhancing their visibility on MRIs
- Imaging marker of Cerebral Small Vessel Disease





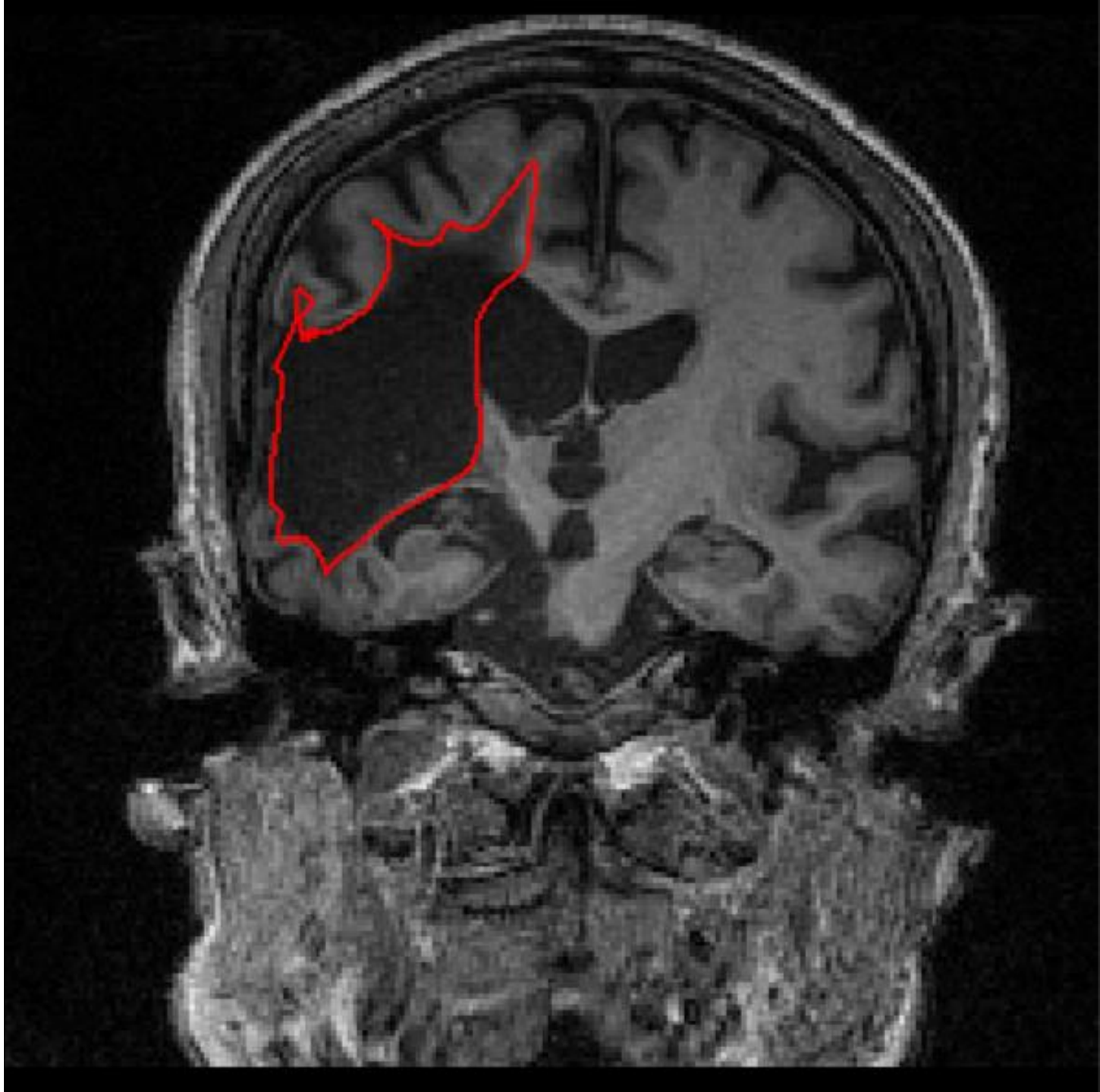
PVS commonly appears in the Basal Ganglia





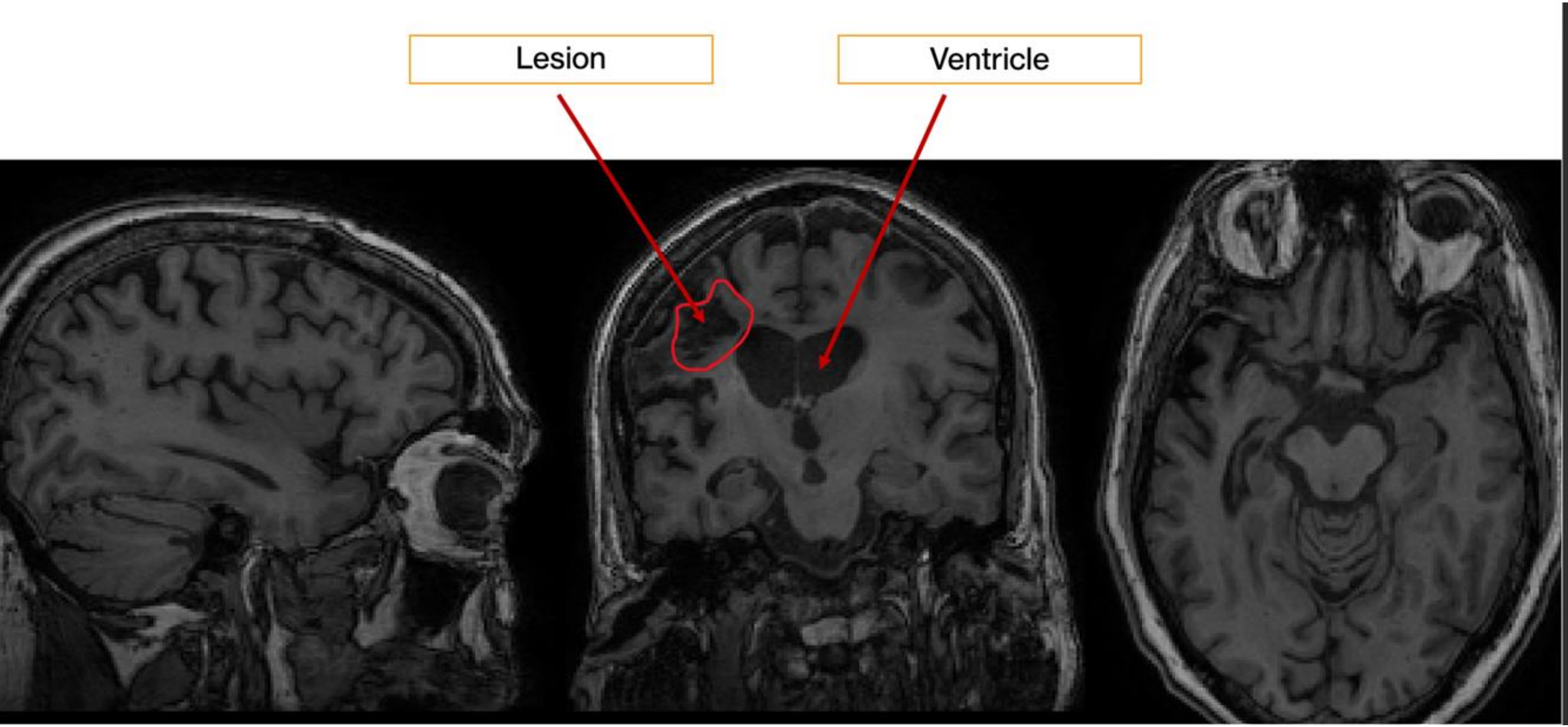
What does a lesion look like?

Border in red



Now that you have identified the ventricles, remember:

- A nearby lesion will impact the ventricles
- In the subject on the next slide, you can see ventricle atrophy next to the lesion


Lesion
Ventricle

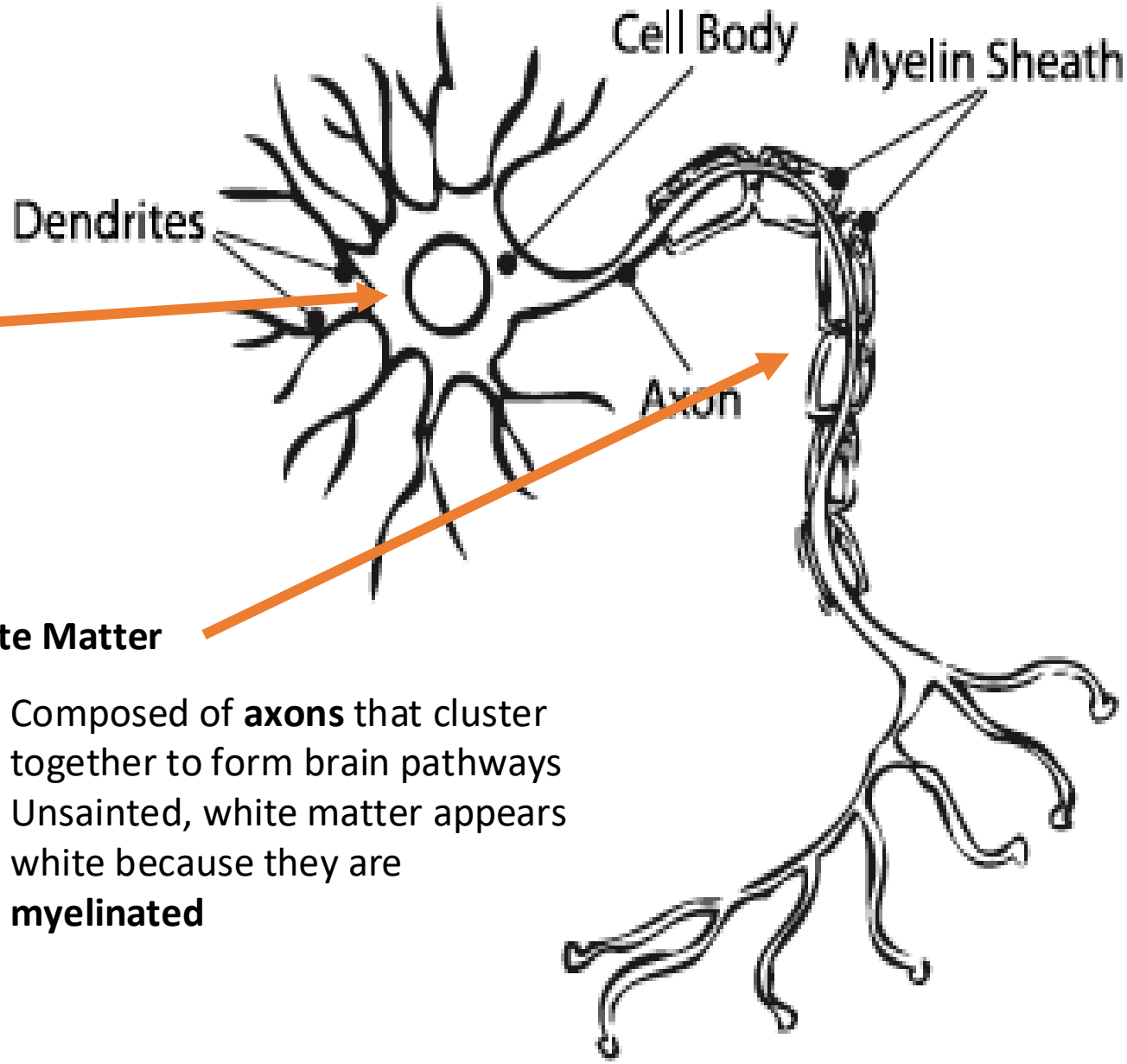
PVS (perivascular spaces) – tricky & appear like "pseudo"lesions, but they are NOT stroke lesions!





MRI Scan Pointers

White Matter vs Gray Matter



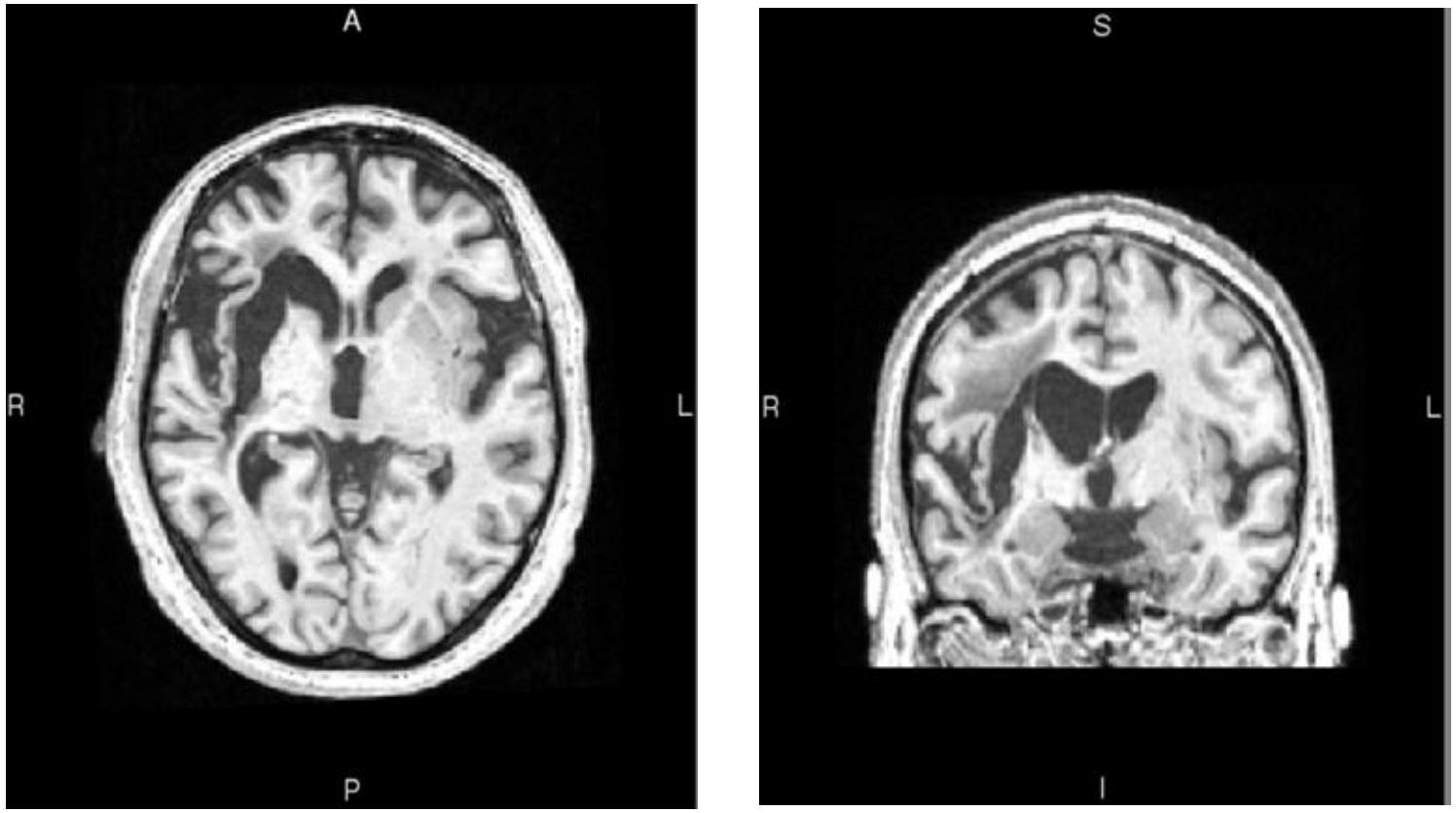
Gray Matter

- Consists mostly of **neuron cell bodies** that cluster together
- Appears “gray” because it contains **very few myelinated axons**

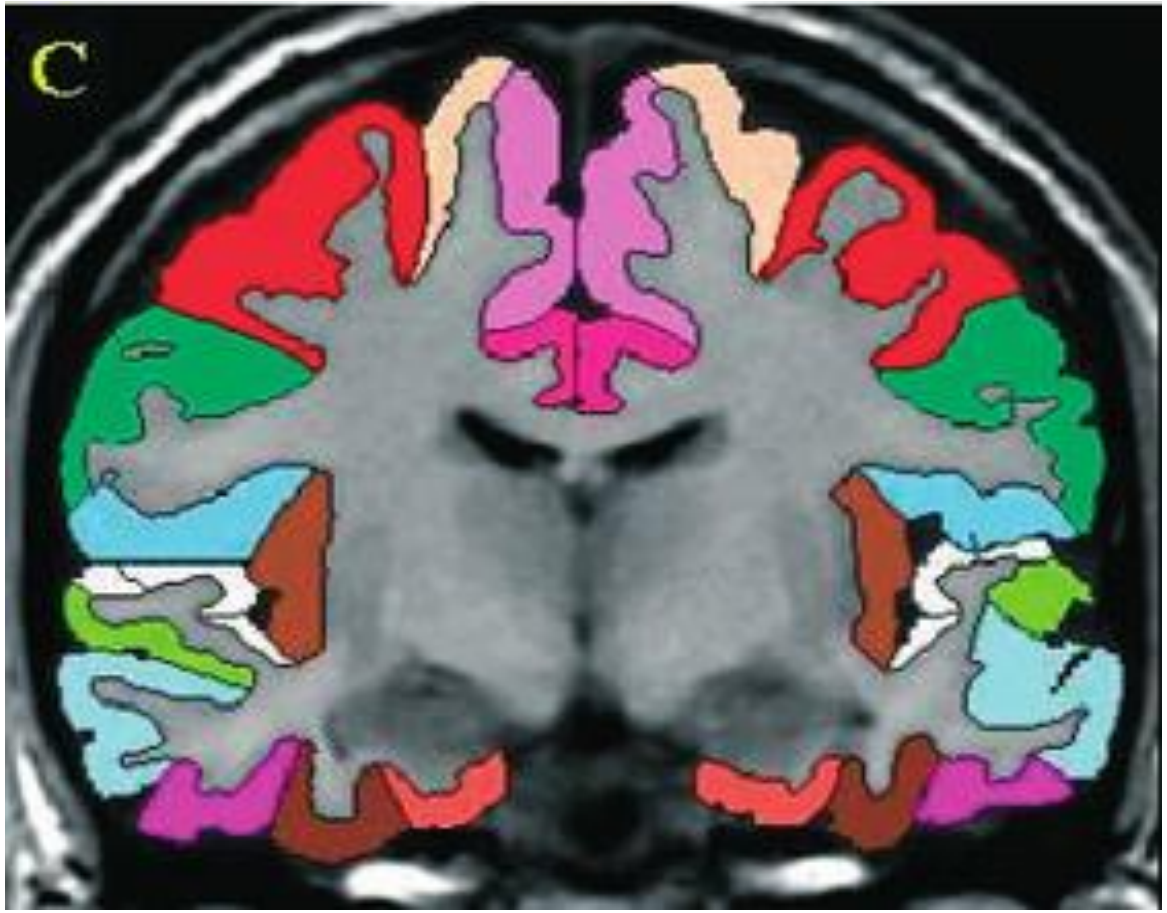
White Matter

- Composed of **axons** that cluster together to form brain pathways
- Unsainted, white matter appears white because they are **myelinated**

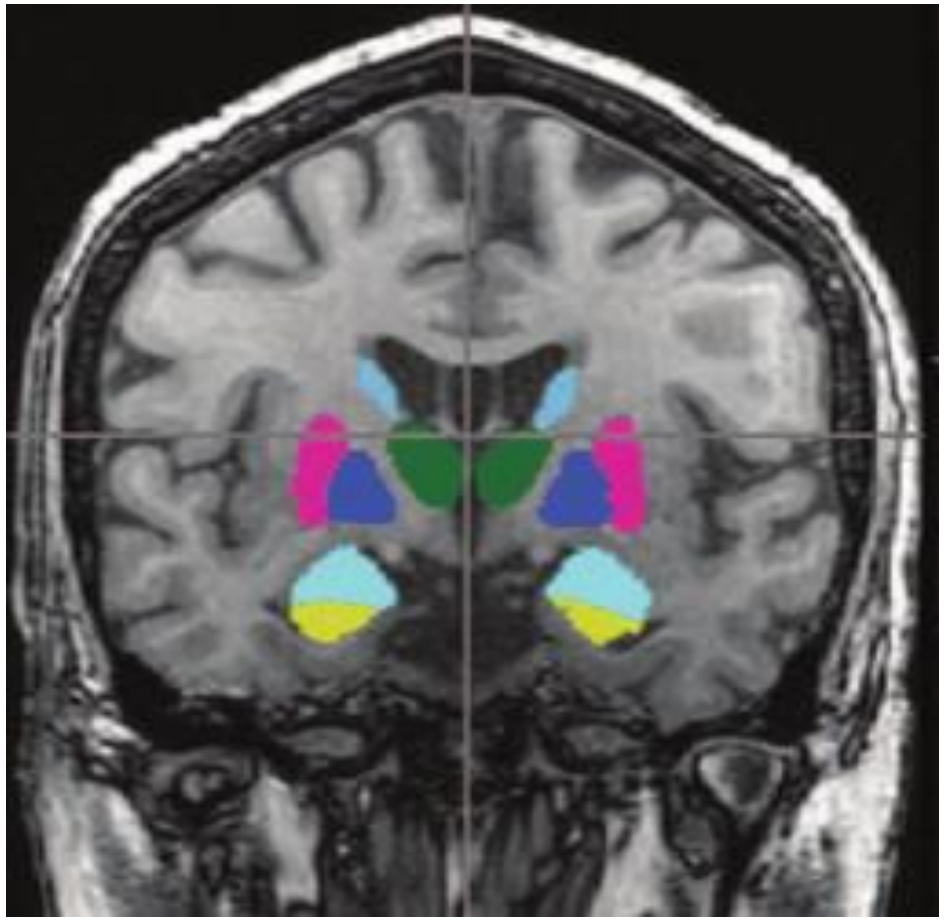
Ventricular Asymmetry- Hydrocephalus Ex Vacuo



Cortical Segmentation

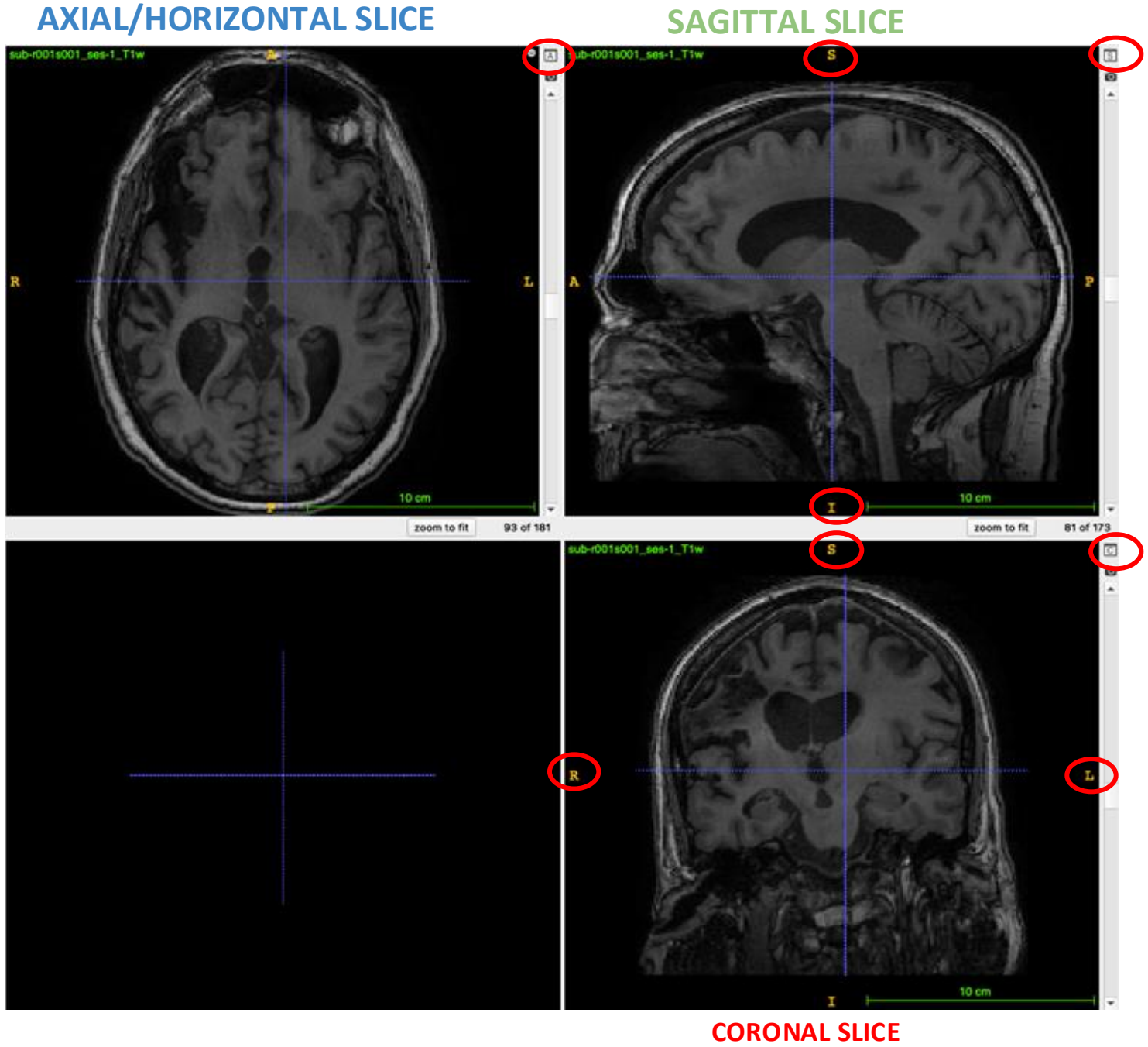


Subcortical Segmentation

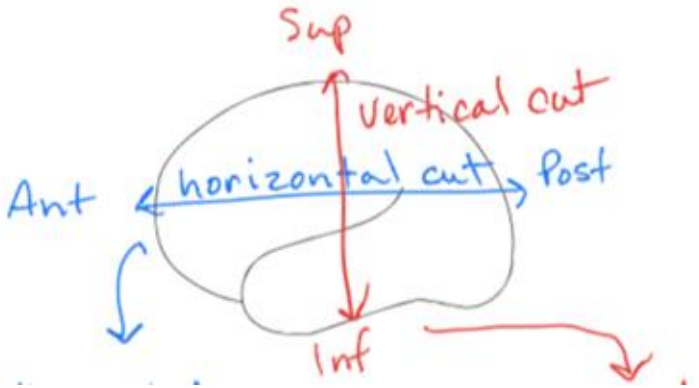


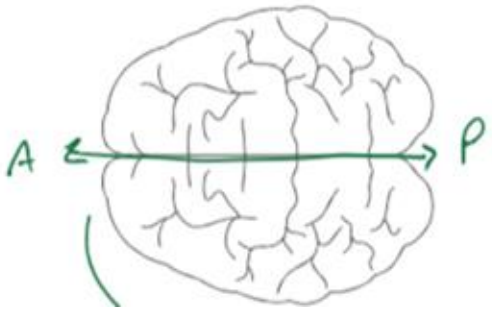
ITK-SNAP

If you open an image & the orientation or anatomical labels are incorrect, don't trace it



Navigating the brain

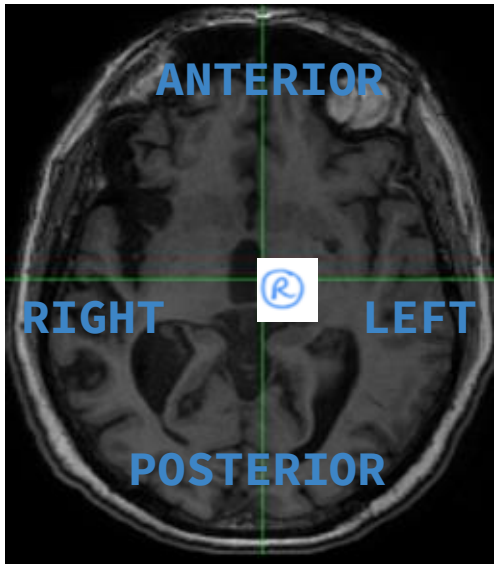




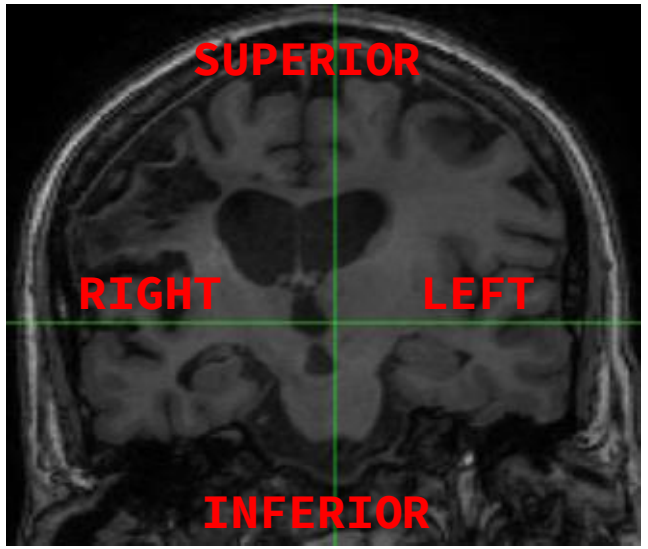
ANTERIOR= ROSTRAL

POSTERIOR= CAUDAL

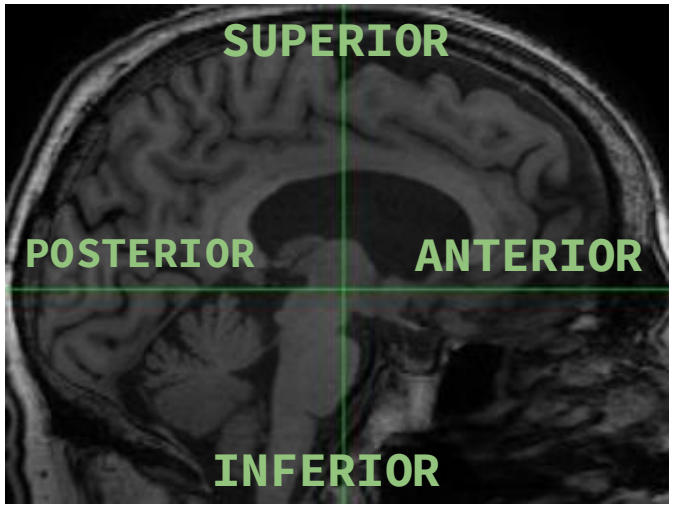
AXIAL/HORIZONTAL SLICE



CORONAL SLICE

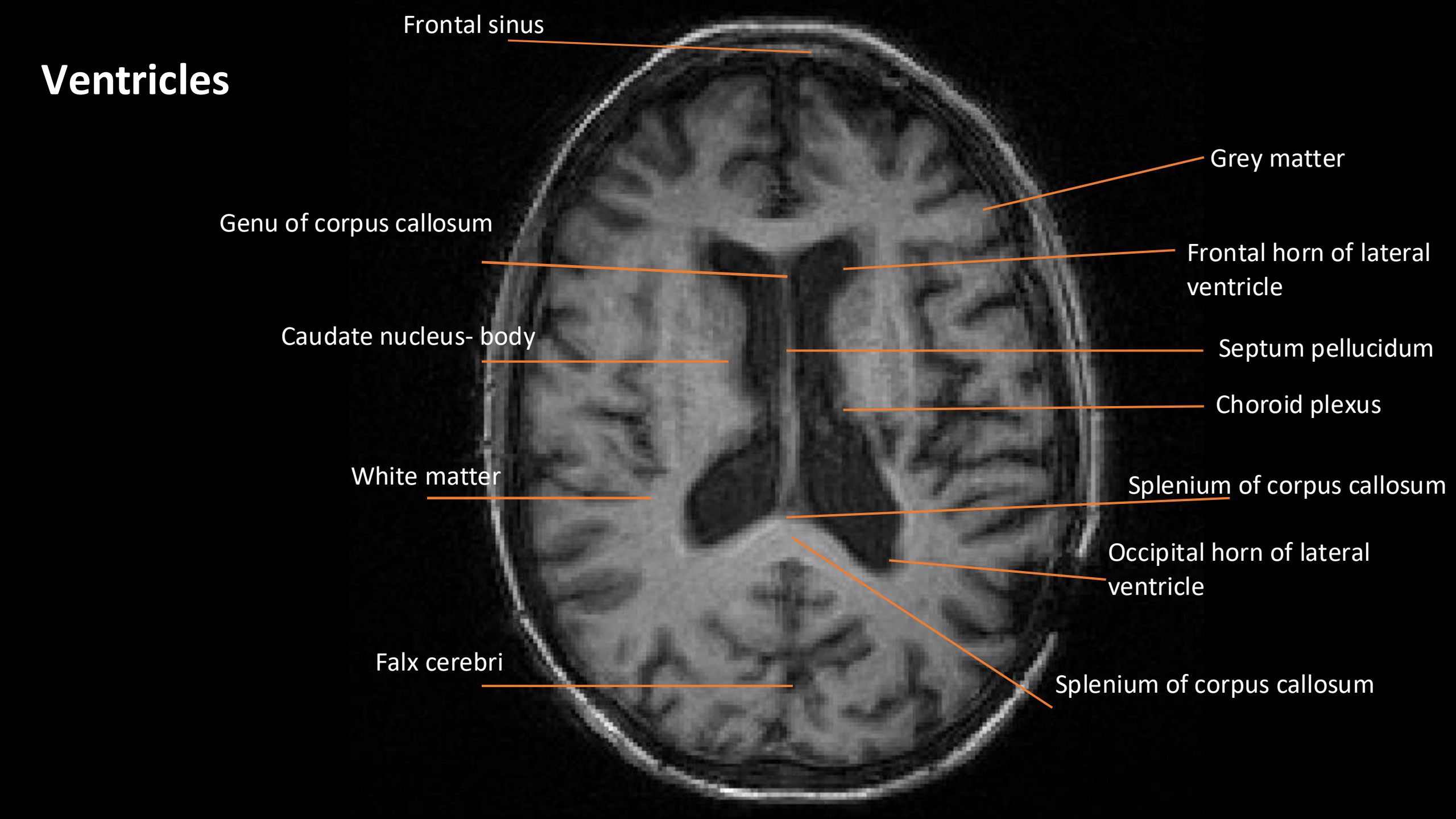


SAGITTAL SLICE

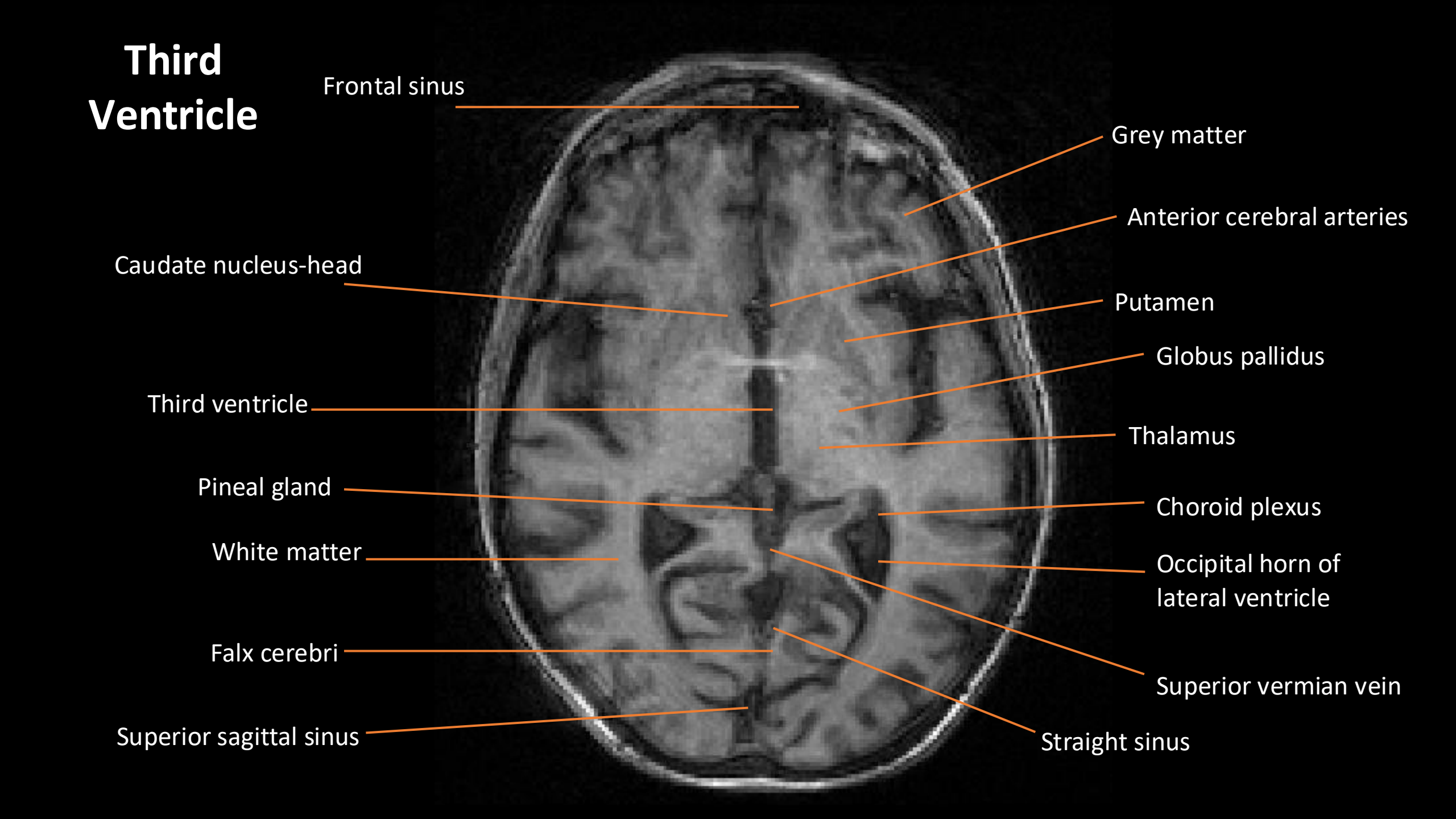


Axial Sections

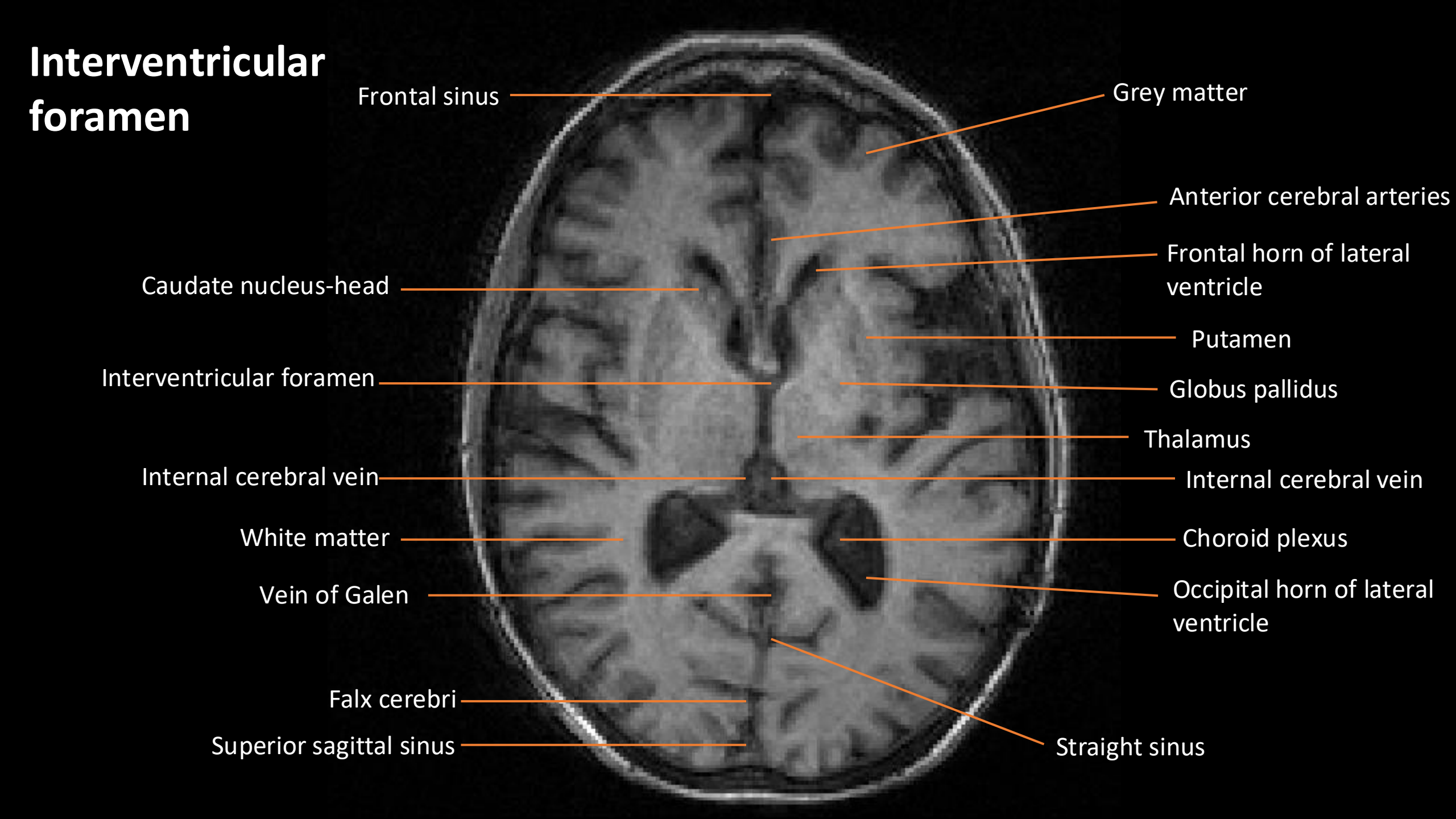
Ventricles


Frontal sinus
Grey matter
Genu of corpus callosum
Frontal horn of lateral ventricle
Caudate nucleus- body
Septum pellucidum
Choroid plexus
White matter
Splenium of corpus callosum
Occipital horn of lateral ventricle
Falx cerebri
Splenium of corpus callosum

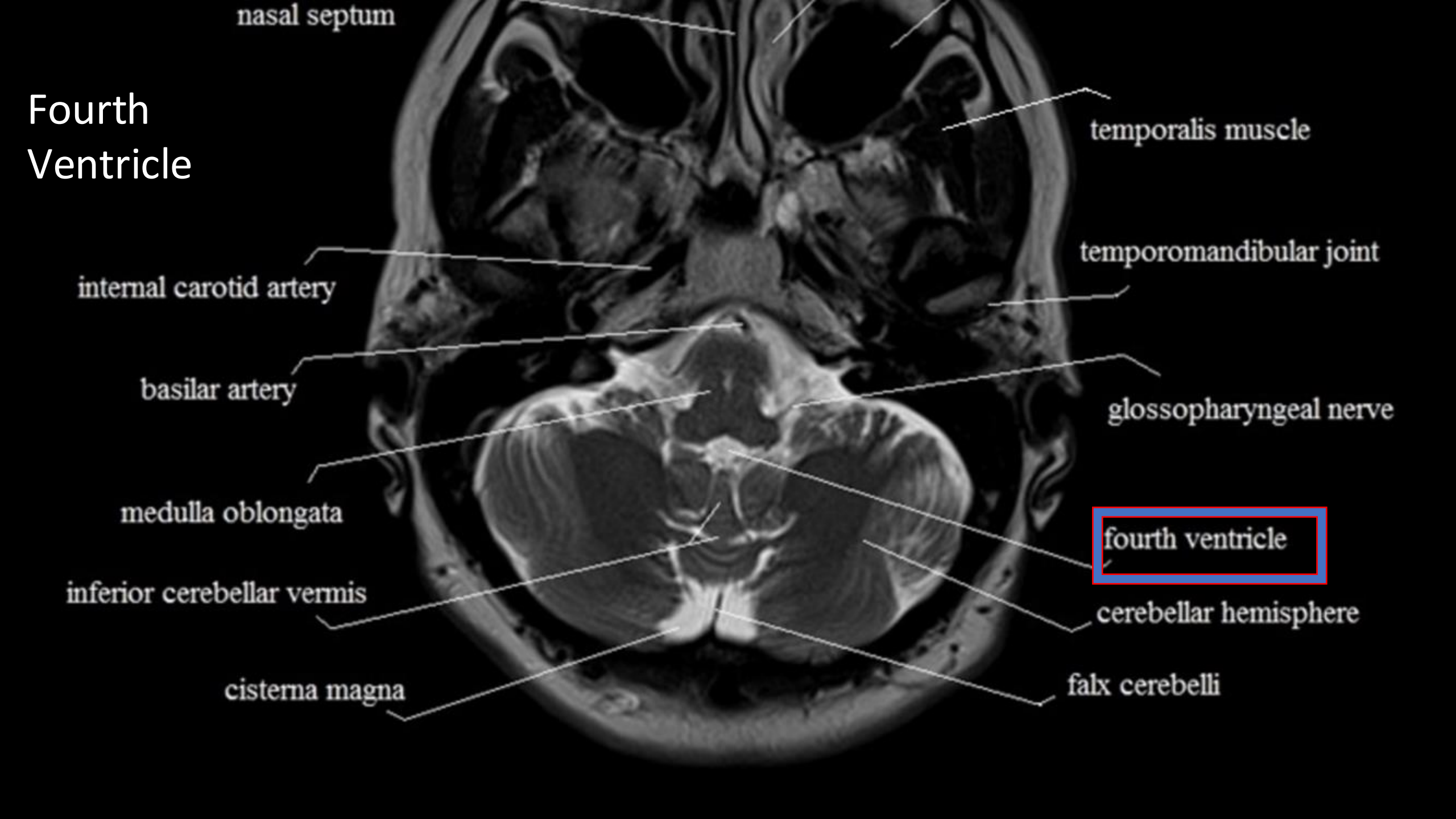
Third Ventricle



Interventricular foramen

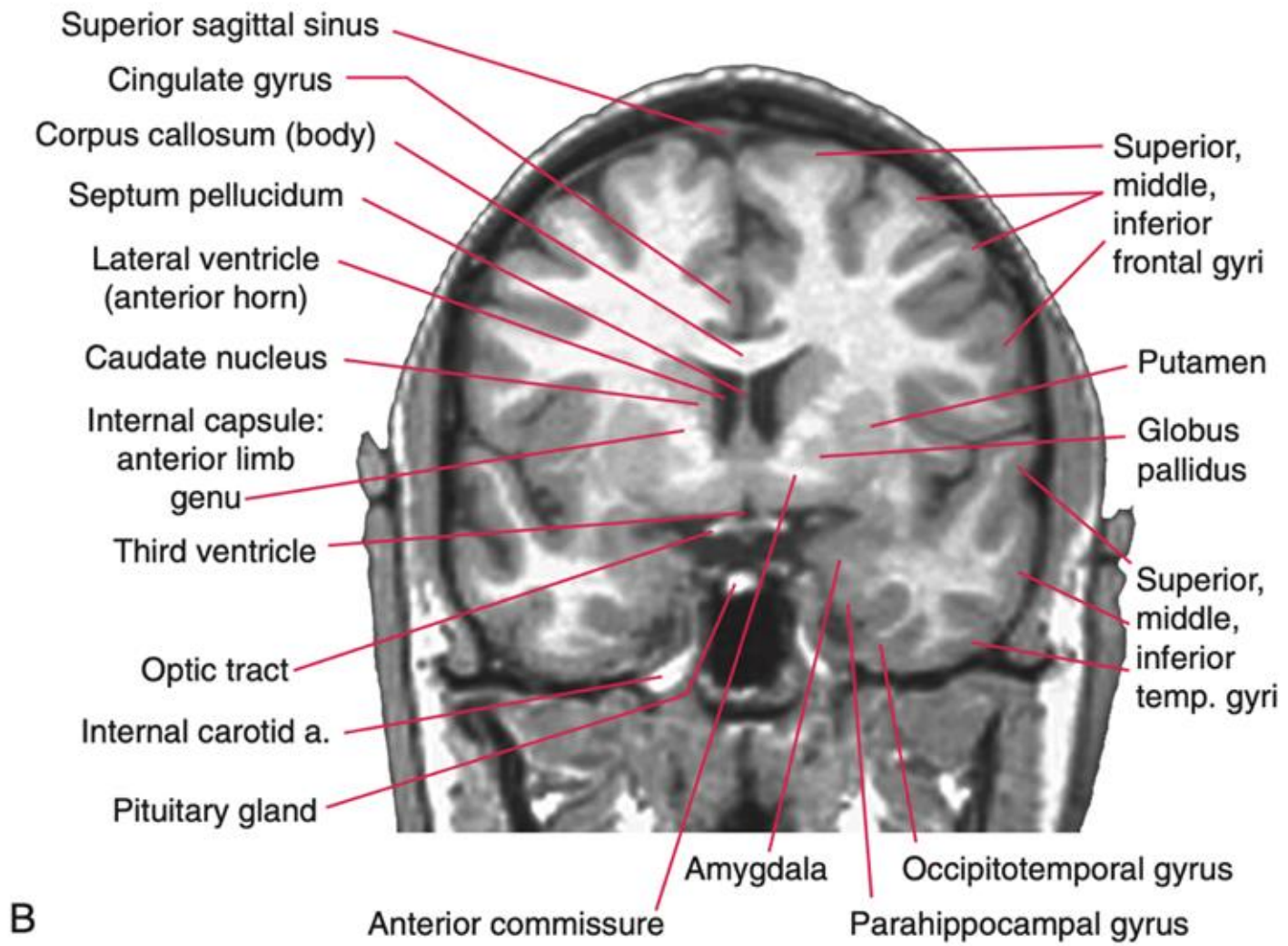


Fourth Ventricle

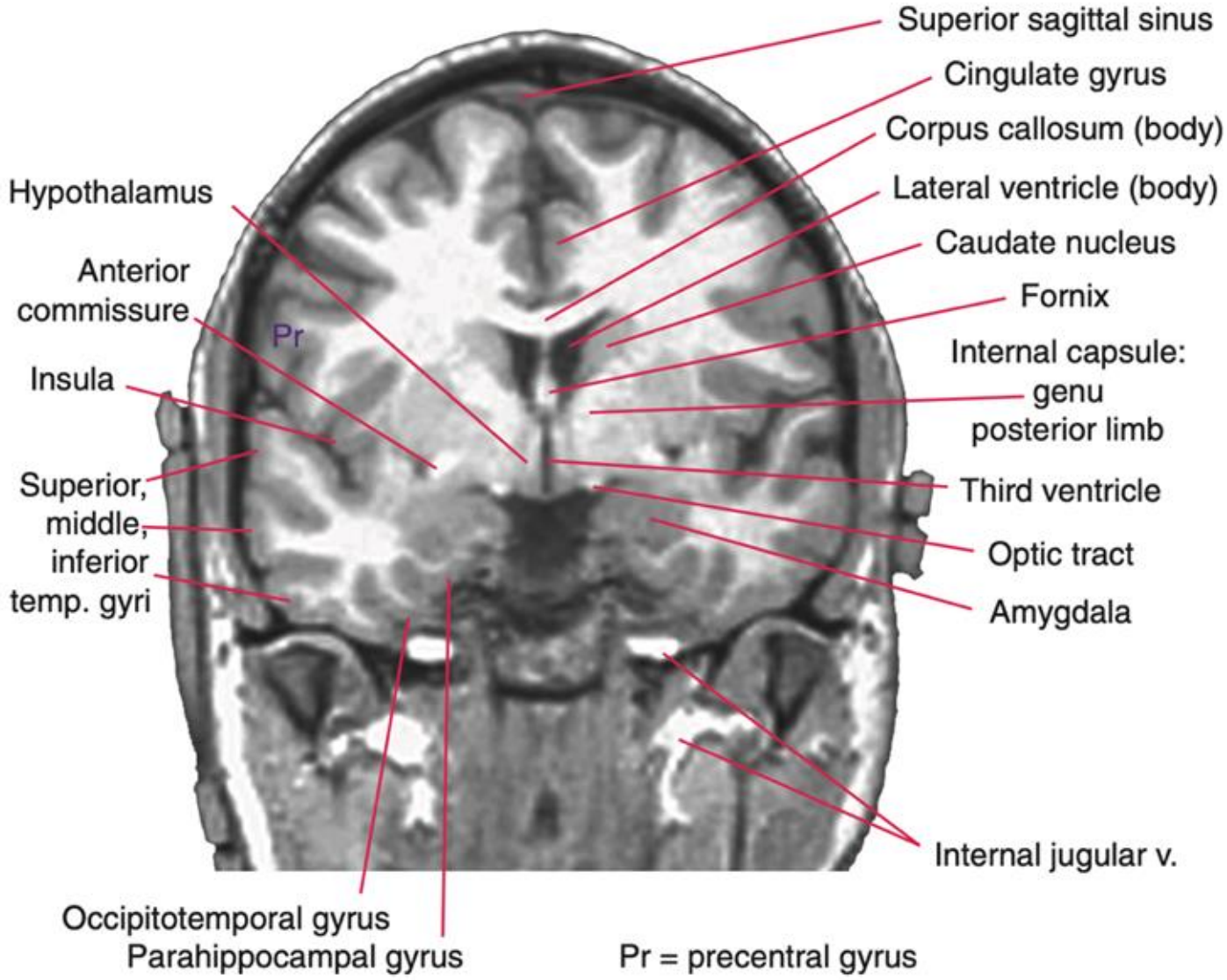


Coronal Sections

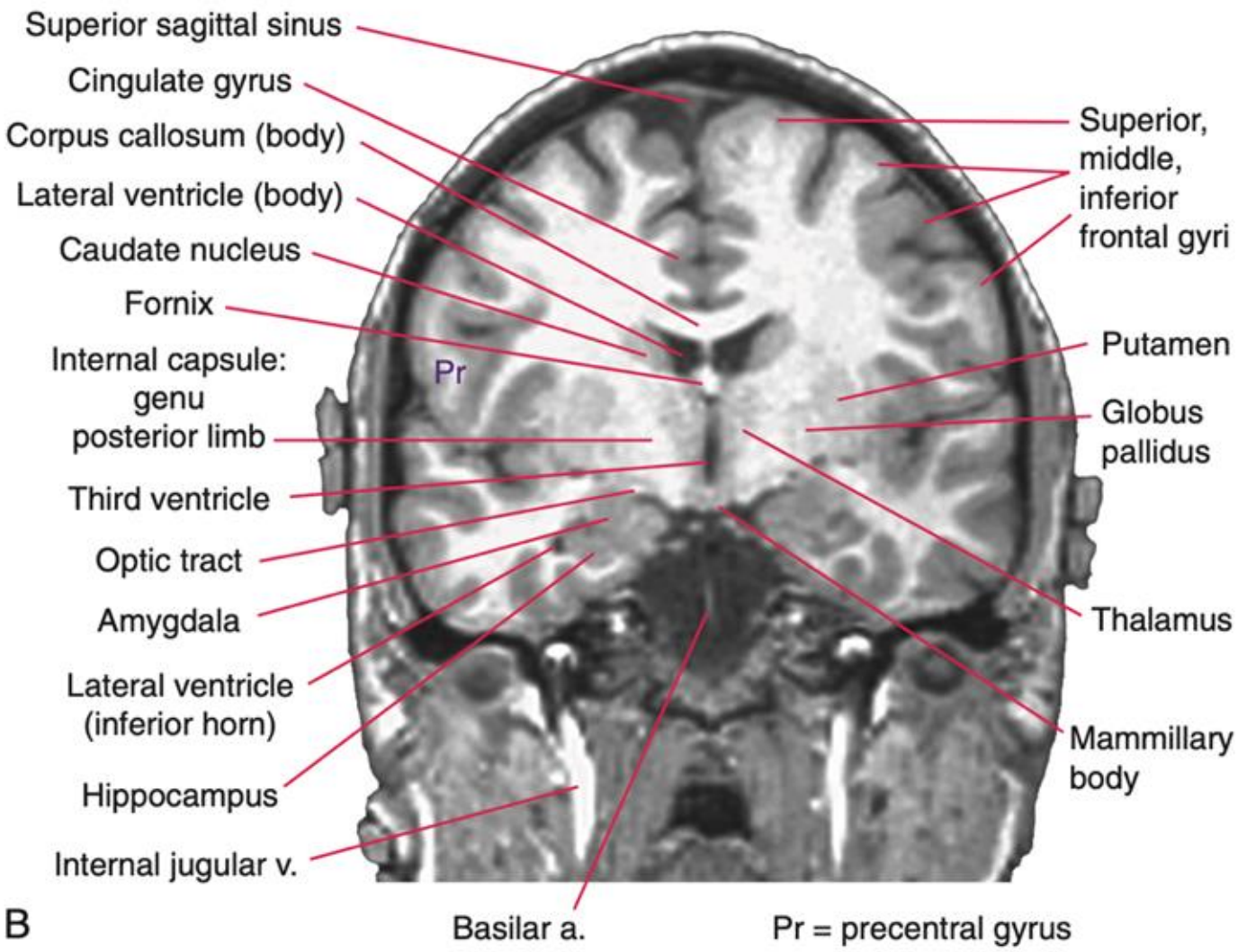
Section through the anterior aspect



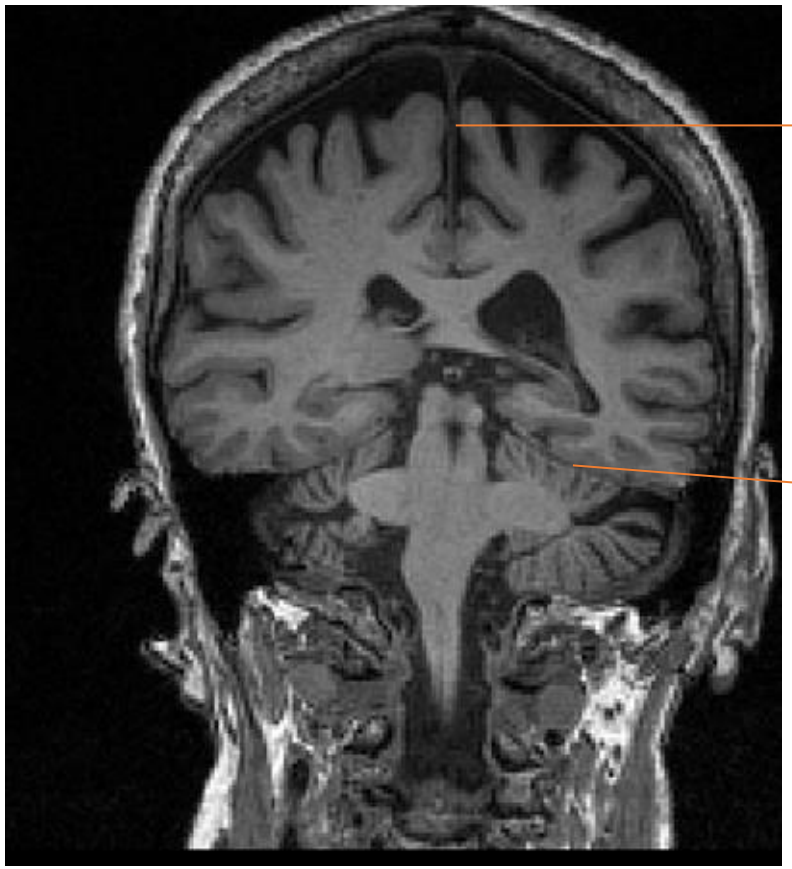
Section through the diencephalon (which includes the thalamus, hypothalamus, and pineal gland)



Section through posterior aspect (thalamus and brainstem)

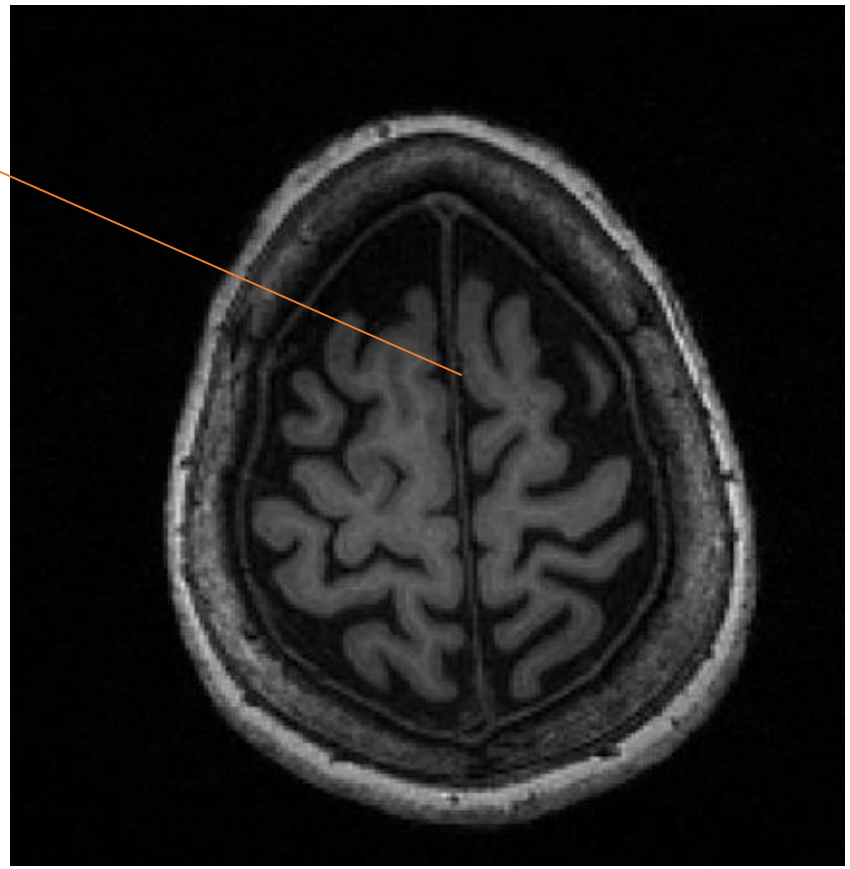


Major dural folds

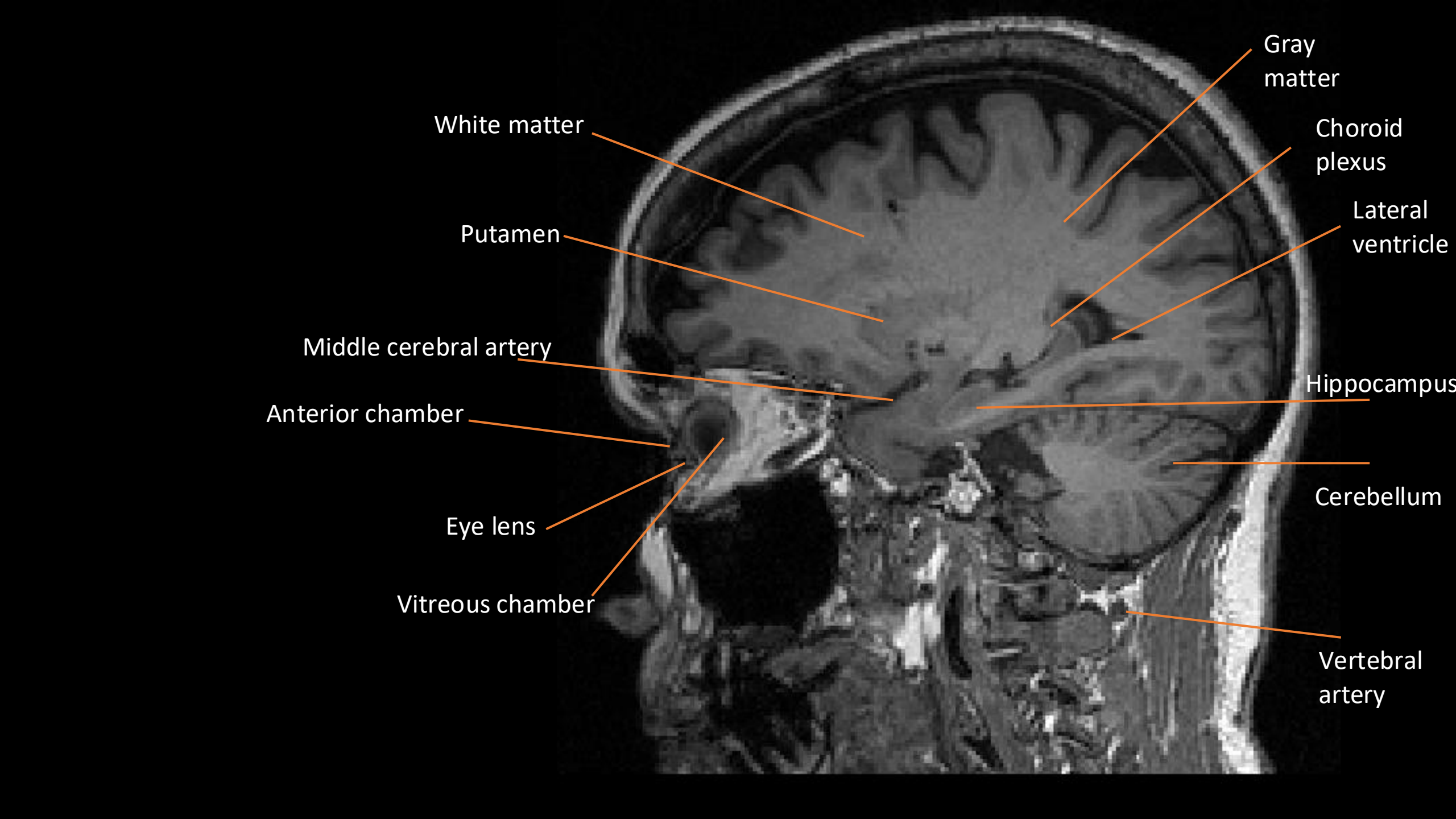


Falx Cerebri

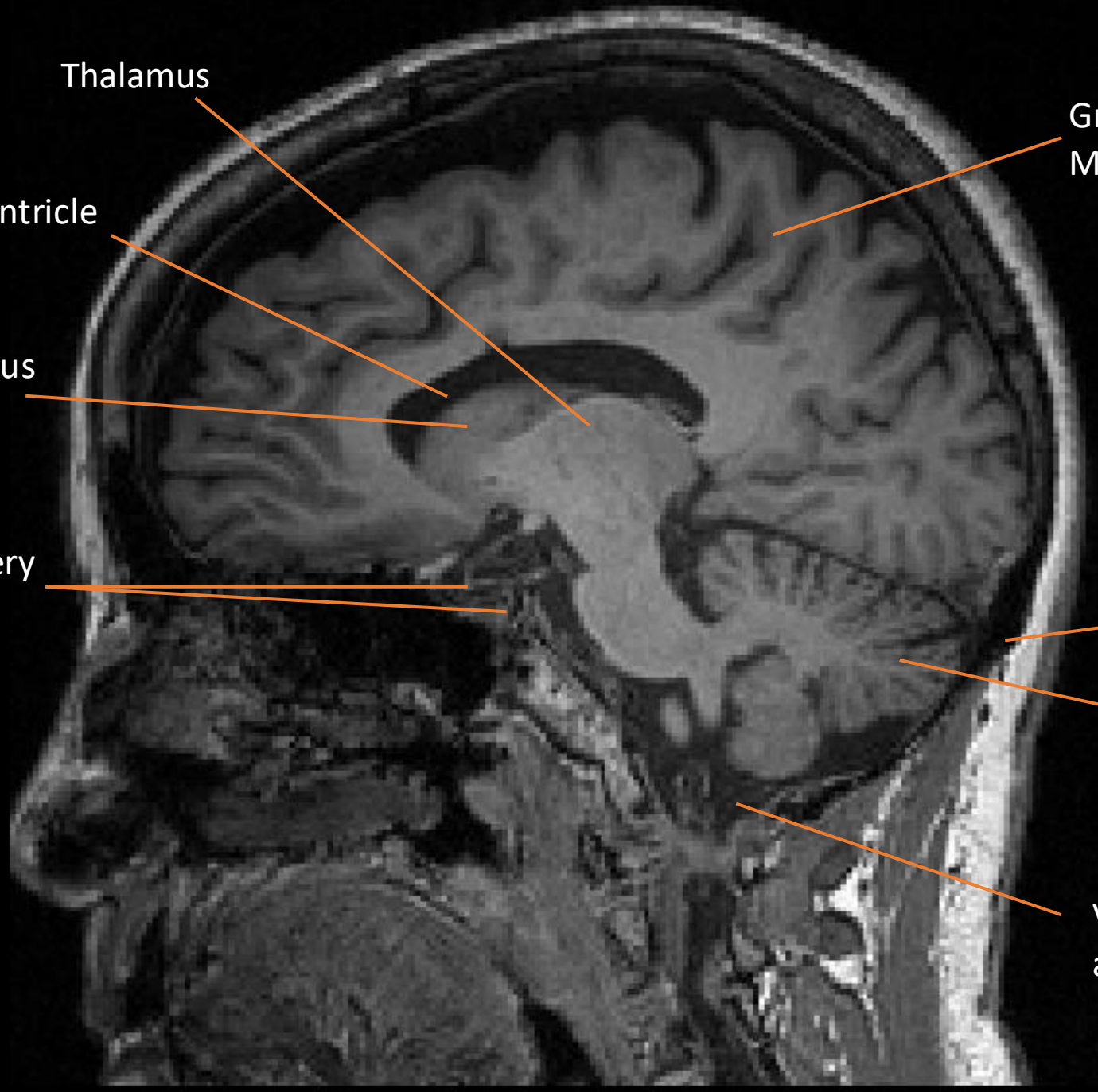
Tentorium cerebelli

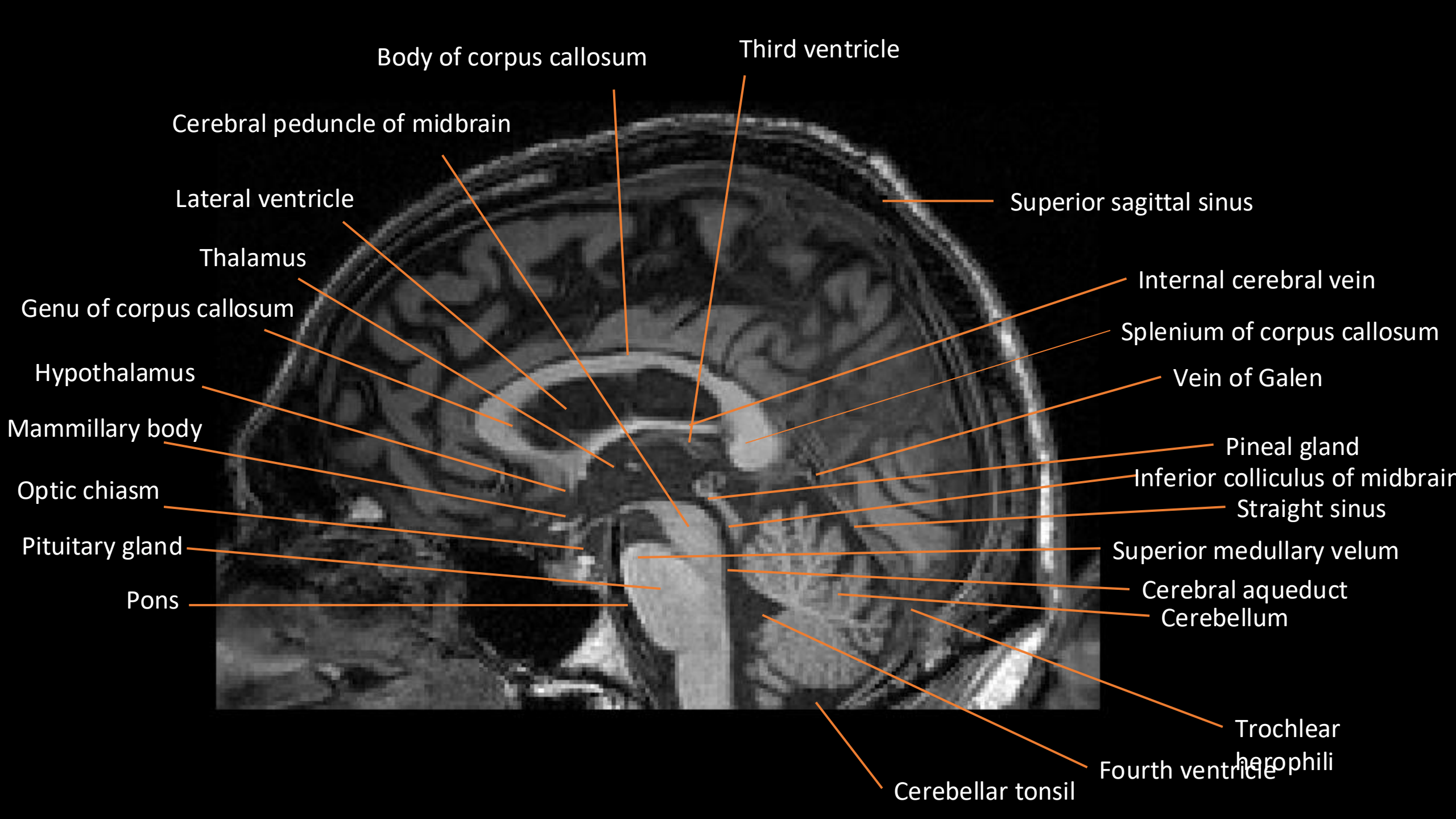


Sagittal Sections


Gray matter
White matter
Choroid plexus
Putamen
Lateral ventricle
Middle cerebral artery
Hippocampus
Anterior chamber
Cerebellum
Eye lens
Vitreous chamber
Vertebral artery

Important midsagittal structures




Body of corpus callosum
Third ventricle
Cerebral peduncle of midbrain
Lateral ventricle
Superior sagittal sinus
Thalamus
Internal cerebral vein
Genu of corpus callosum
Splenium of corpus callosum
Hypothalamus
Vein of Galen
Mammillary body
Pineal gland
Optic chiasm
Inferior colliculus of midbrain
Straight sinus
Pituitary gland
Superior medullary velum
Pons
Cerebral aqueduct
Cerebellum
Trochlear
Fourth ventricle
Cerebellar tonsil